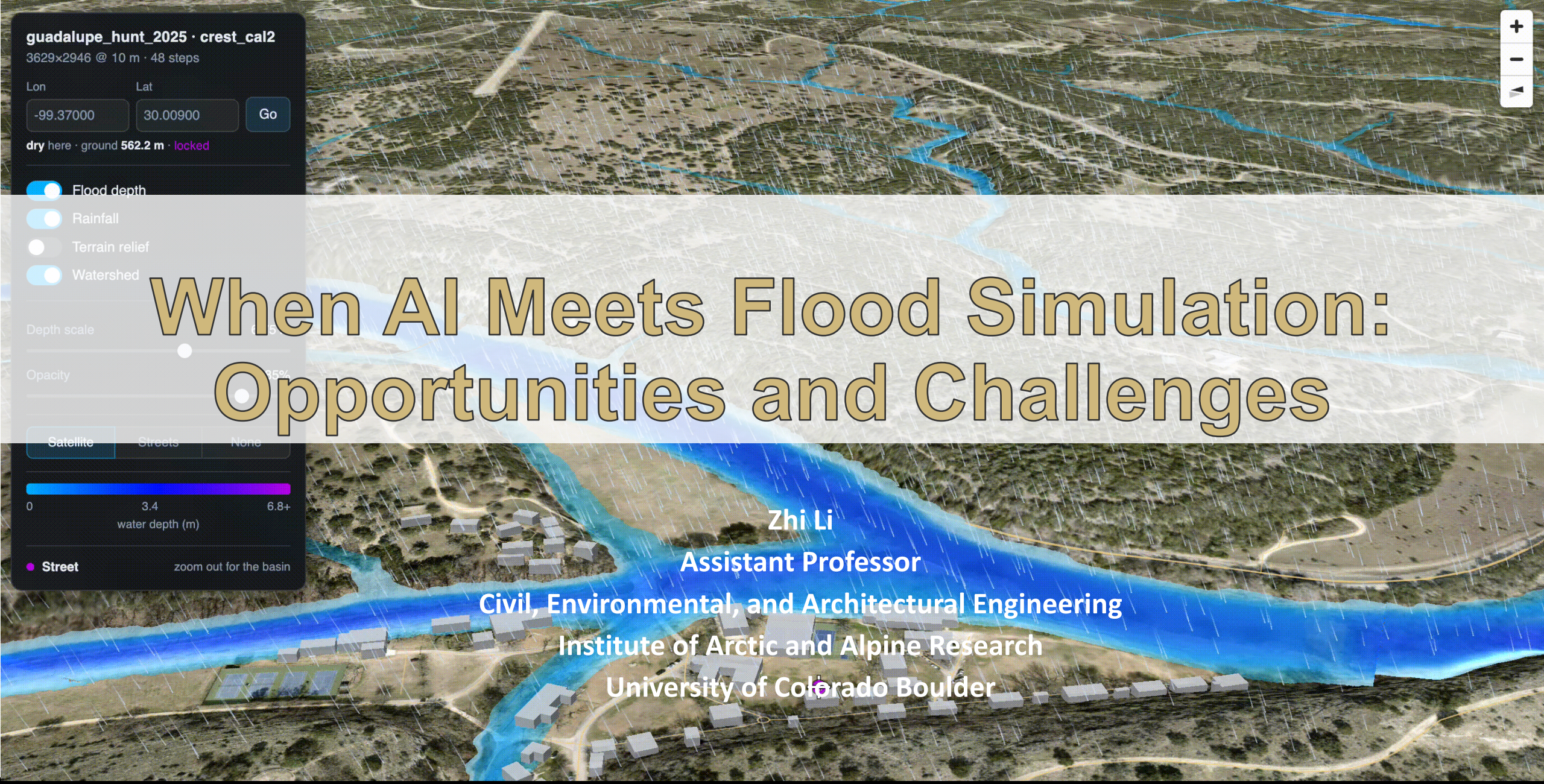


When AI Meets Flood Simulation: Opportunities and Challenges

Zhi Li

Assistant Professor

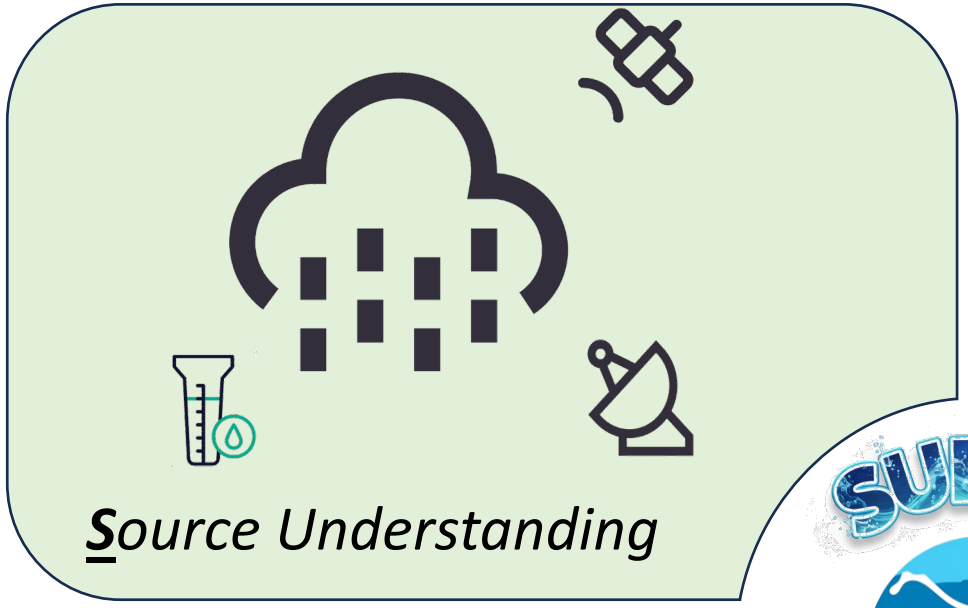
Civil, Environmental, and Architectural Engineering

Institute of Arctic and Alpine Research

University of Colorado Boulder



https://hydrors.us



SURGE

Flood Lab

https://hydrors.us

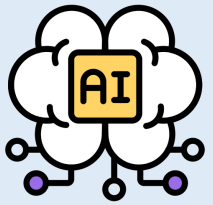


***S**ource Understanding*

SURGE

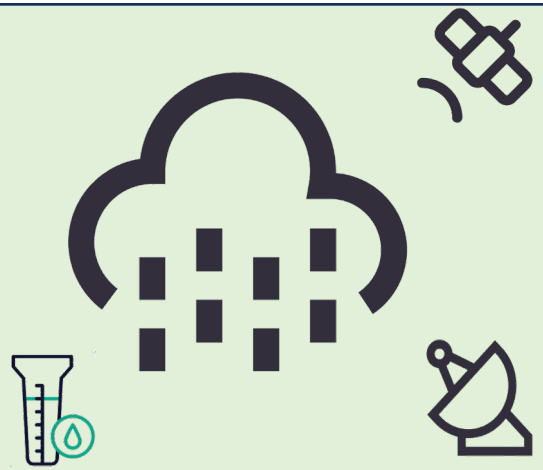
Flood Lab





***U**nifying Models*

https://hydrors.us



***S**ource Understanding*

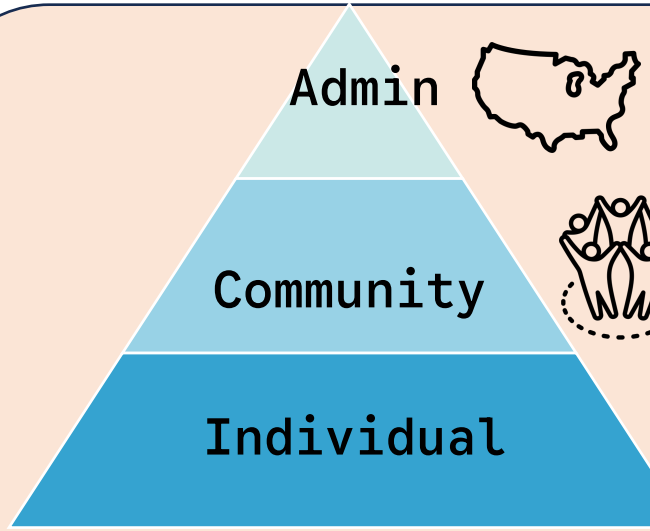
SURGE

Flood Lab



***U**nifying Models*

https://hydrors.us



***R**isk Communication*



***S**ource Understanding*

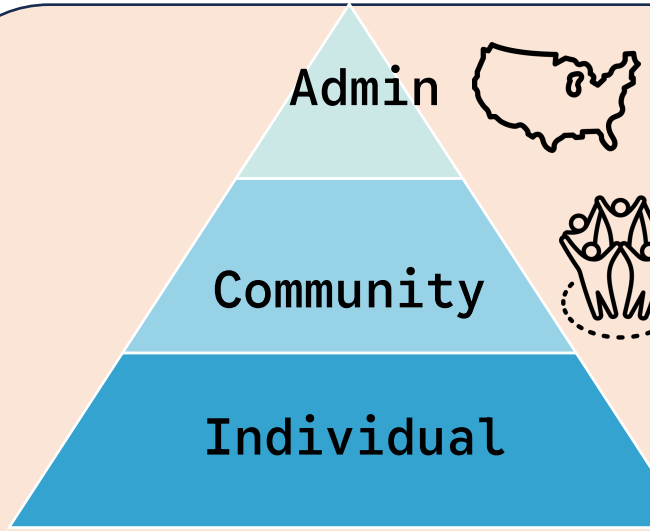
SURGE

Flood Lab



***U**nifying Models*

https://hydrors.us



***R**isk Communication*



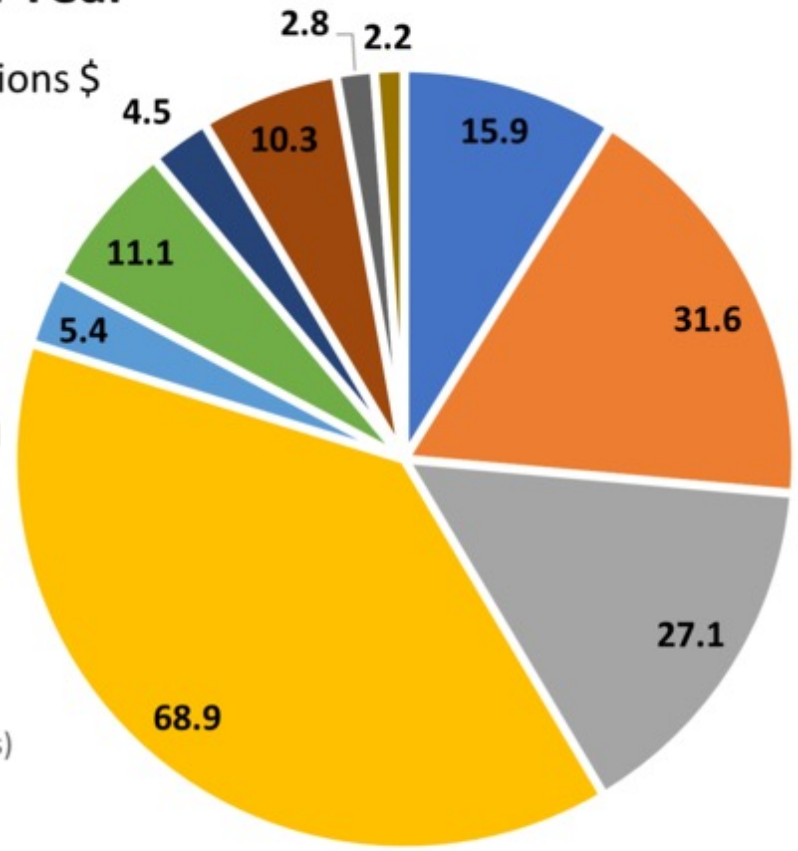
***G**lobal & Sectoral **E**valuation*

Floods are the most Damaging Natural Hazards

Floods Cost Over $180 Billion a Year

Annual Total Costs and Losses in 2023 Billions $

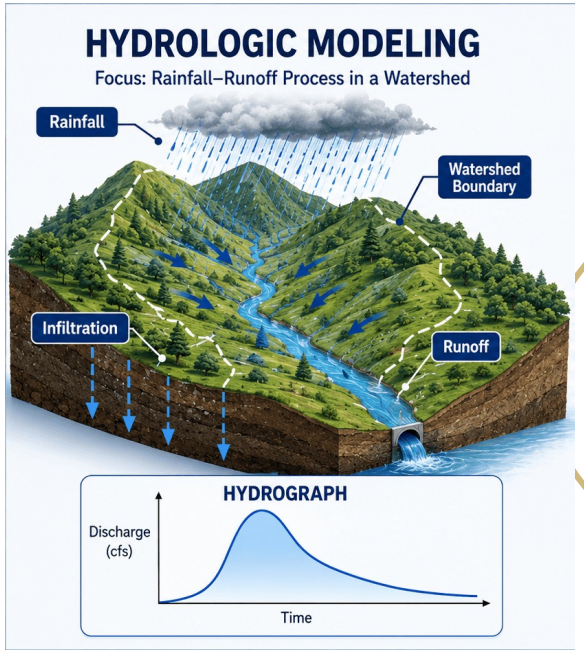
- Structural damage to commercial physical assets
- Direct commercial impact (direct lost output)
- Indirect commercial impact (due to downtime days)
- Infrastructure needs for solutions
- Total value of owned outright homes lost to sea level rise
- Expected annual damage to homes with federally backed mortgages
- EAD to non-federally backed homes
- Decreased tax revenue
- Insurance costs (claims, crop loss, premium increases)
- Other



Note: Other costs include school infrastructure damage (only for WV), damage to ecosystem services, transit infrastructure damage (only for Boston MBTA), and deaths

JEC

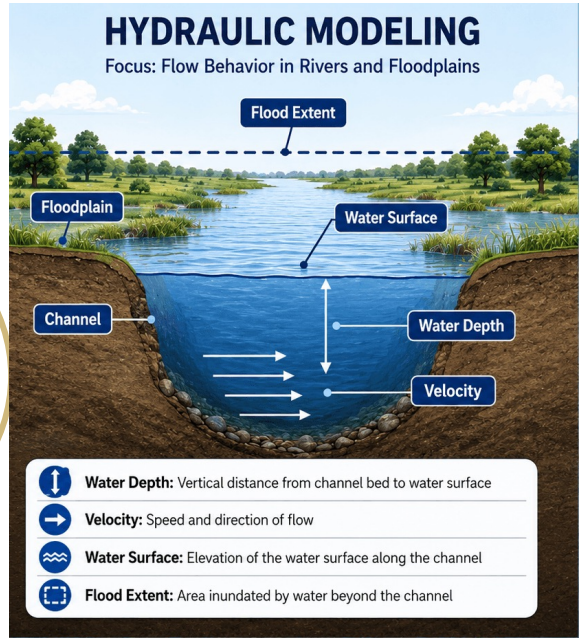
What is a flood model?



Hydrologic model

Hydraulic model

Hydrologic-Hydraulic (H&H) model



Contents

- Review of 50 years of flood model development
- Physics + AI flood modeling
- Agent-based flood simulation
- Prospect: Opportunities and challenges

Flashback: what have we done?

Li, Z., & Gorelick, S. M. (2025). Societal and environmental interconnections: future directions for flood inundation models. *Environmental Research Letters*, *20*(12), 123004.

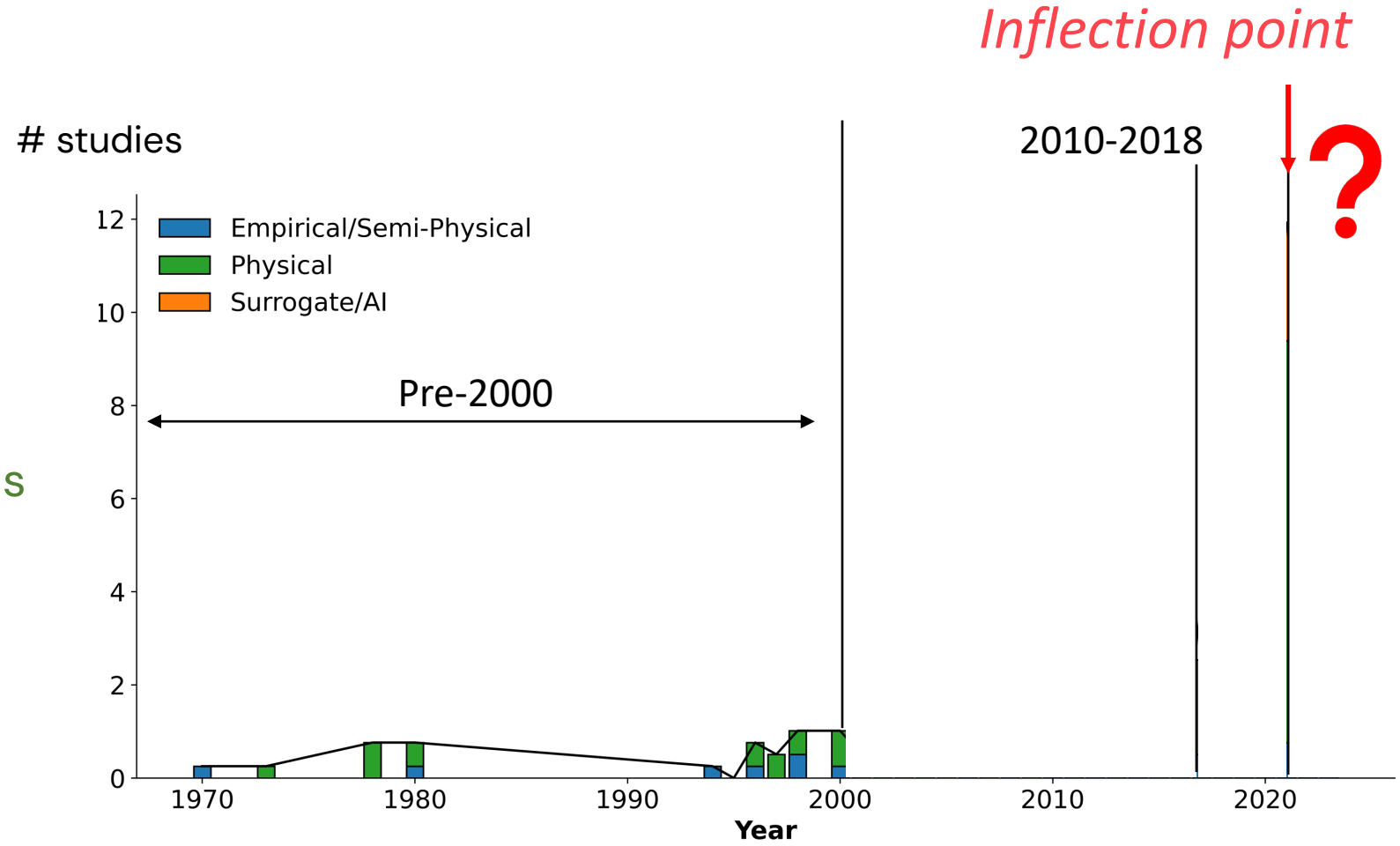
Over the Landscape of Flood Model Research

479 studies from 1970 to 2023

Pre – 2000: Empirical and physical models

2000–2018: Surge in physical models

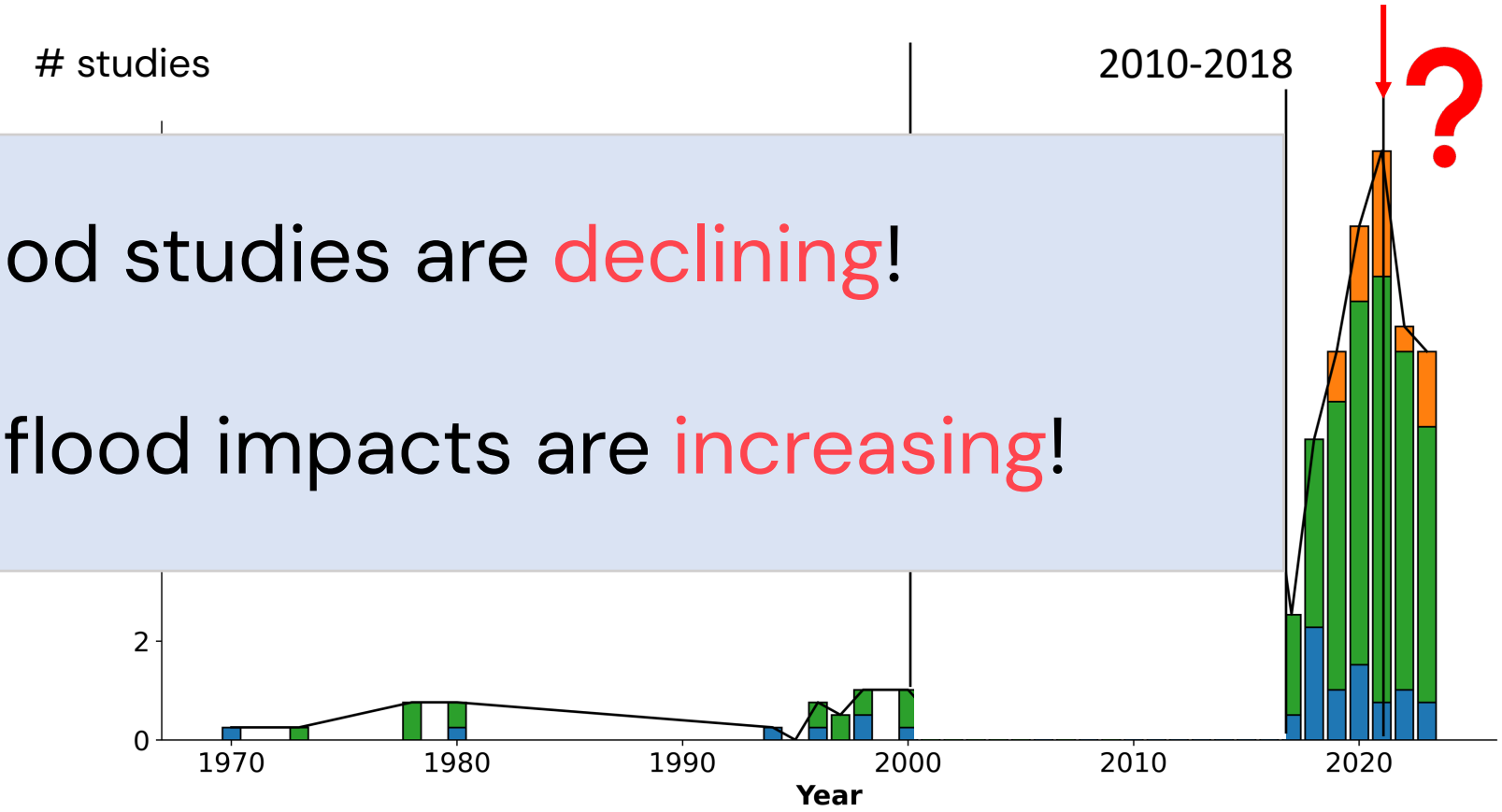
2018–2021: Flourishing period for AI



Li et al., 2025, ERL

Over the Landscape of Flood Model Research

479 studies from 1970 to 2023



Number of flood studies are declining!

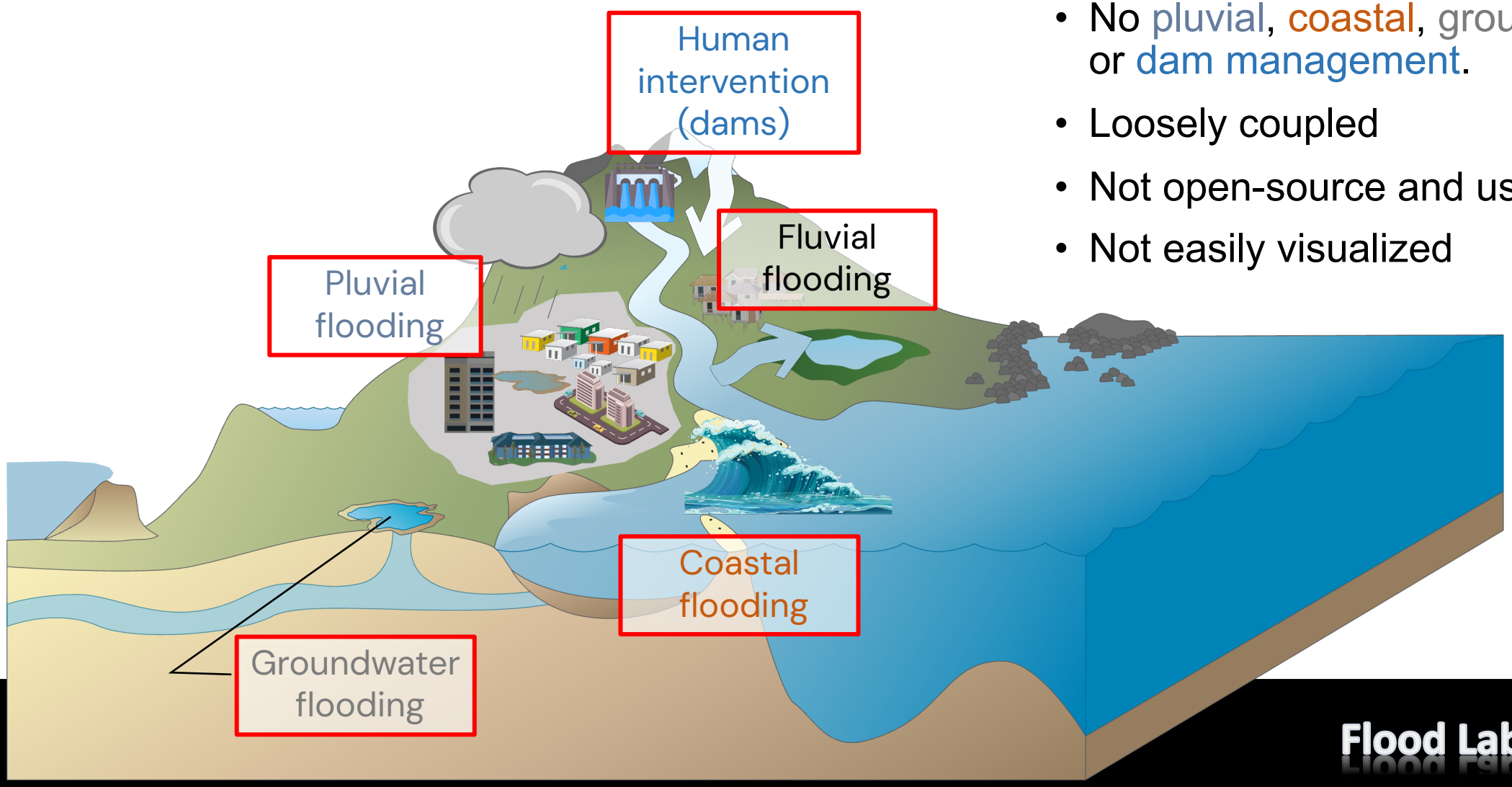
At the same time, flood impacts are increasing!

2018-2021. Flourishing period for AI

Li et al., 2025, ERL

Limitations of Current Flood Inundation Modeling

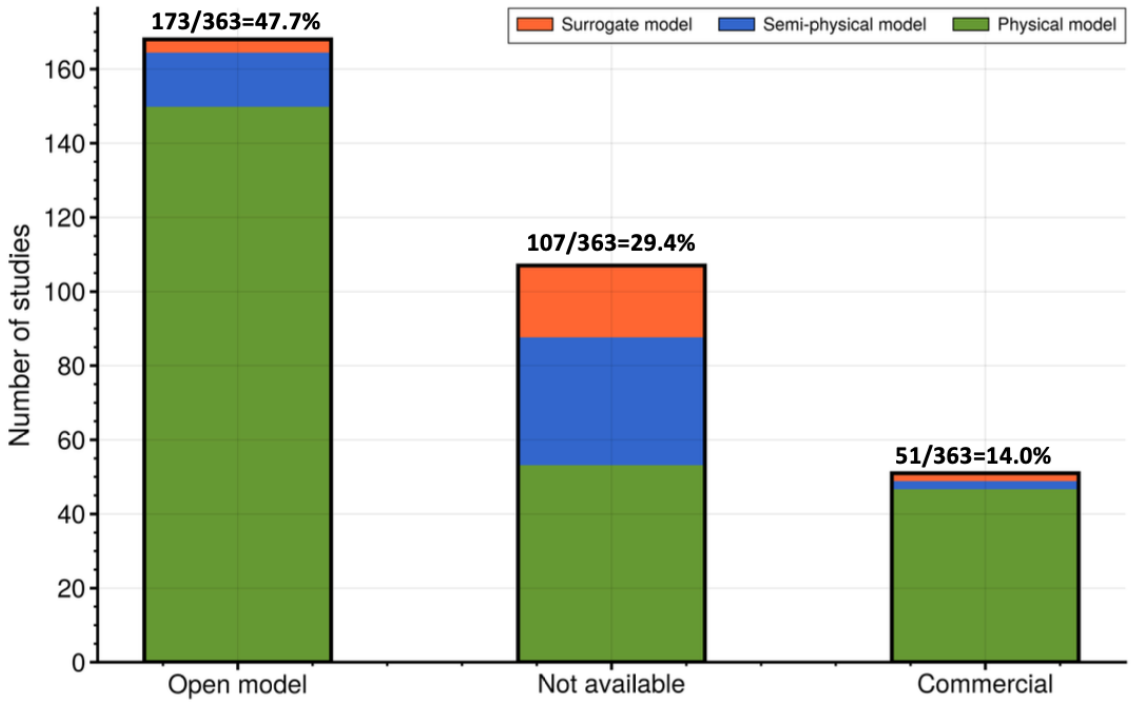
- Only river flooding
- No pluvial, coastal, groundwater floods, or dam management.
- Loosely coupled
- Not open-source and user-unfriendly
- Not easily visualized



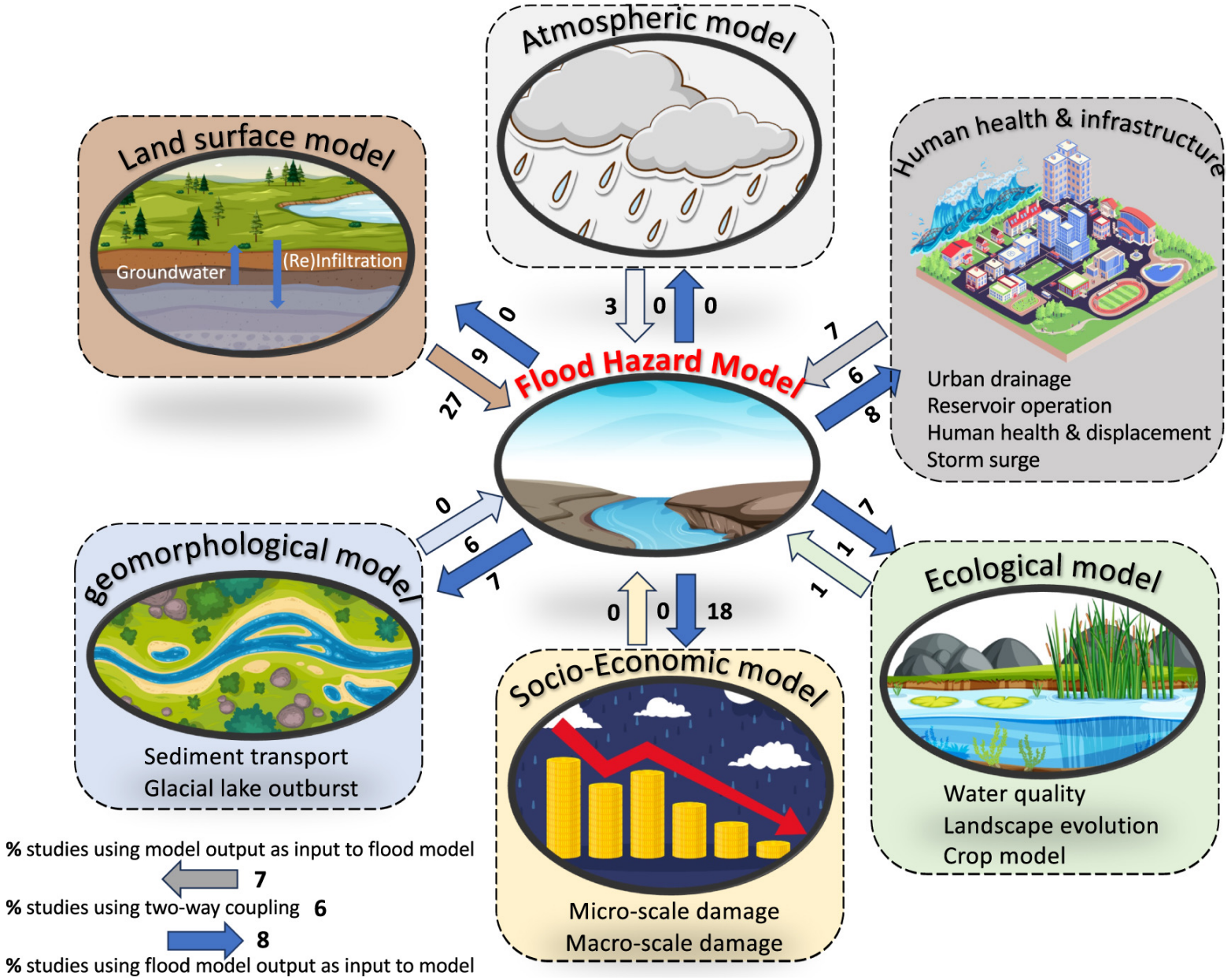
Li et al., 2025, ERL

AI-based Flood Models Are Fast, But

- Not generalizable
- Not trustworthy
- Not open



Future model coupling

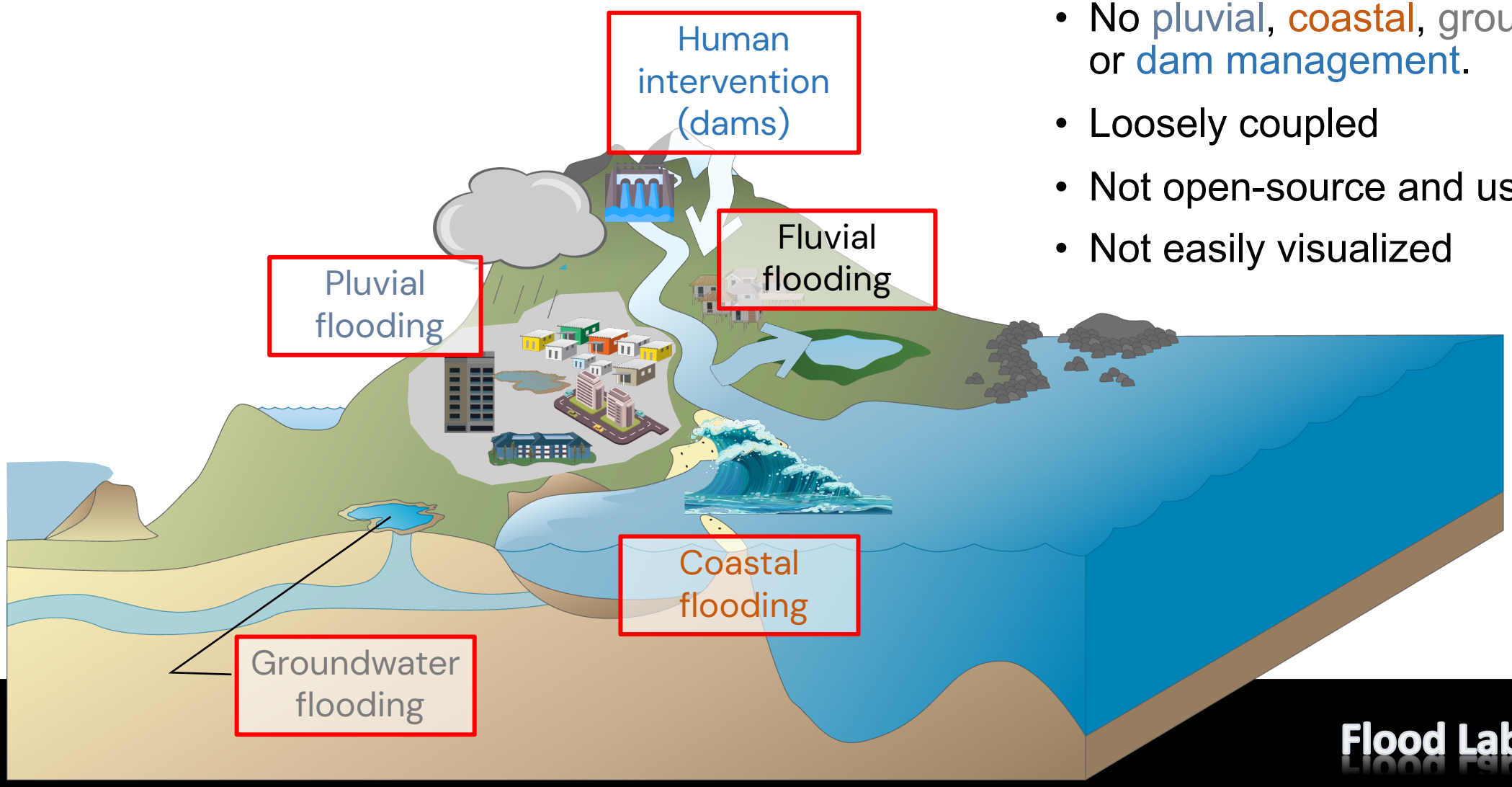


Physics + AI based flood inundation model

Li, Z., Chen, M., Gao, S., Luo, X., Gourley, J. J., Kirstetter, P., ... & Hong, Y. (2021). CREST-iMAP v1. 0: A fully coupled hydrologic-hydraulic modeling framework dedicated to flood inundation mapping and prediction. *Environmental Modelling & Software*, *141*, 105051.
Chen, M., Li, Z., Gao, S., Luo, X., Wing, O. E., Shen, X., ... & Hong, Y. (2021). A comprehensive flood inundation mapping for Hurricane Harvey using an integrated hydrological and hydraulic model. *Journal of Hydrometeorology*, *22*(7), 1713-1726.
Li, Z. (2026). Inunda: A GPU-Native, Agent-enabled, Differentiable Solver for High-Resolution Flood Inundation Modeling. *ArXiv*. https://arxiv.org/abs/2607.09614

Limitations of Current Flood Inundation Modeling

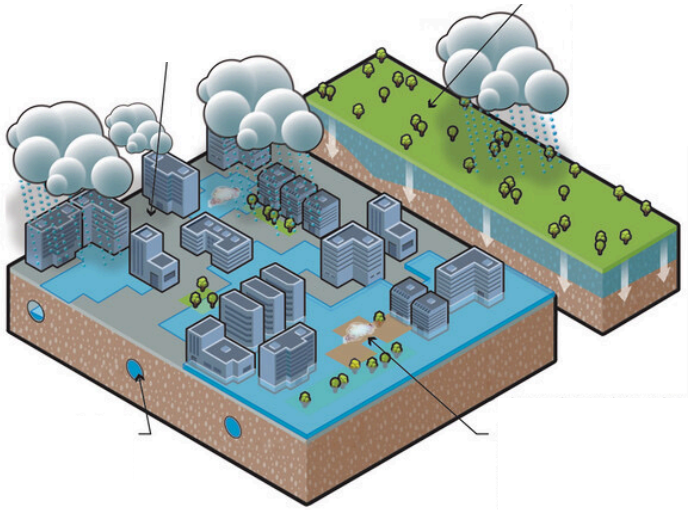
- Only river flooding
- No pluvial, coastal, groundwater floods, or dam management.
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- Not open-source and user-unfriendly
- Not easily visualized



Li et al., 2025, ERL

Developed CREST-iMAP to Overcome These Limitations

Pluvial Flood

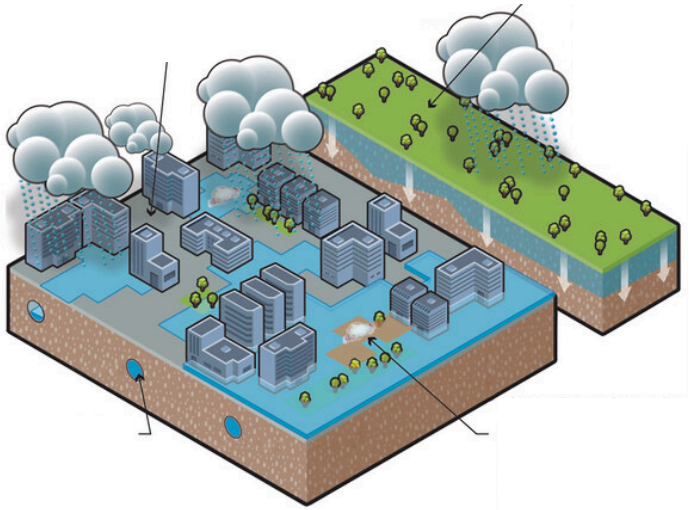




Li et al., 2021, 2022, Environ. Model. Softw.

Developed CREST-iMAP to Overcome These Limitations

Pluvial Flood



CREST-iMAP

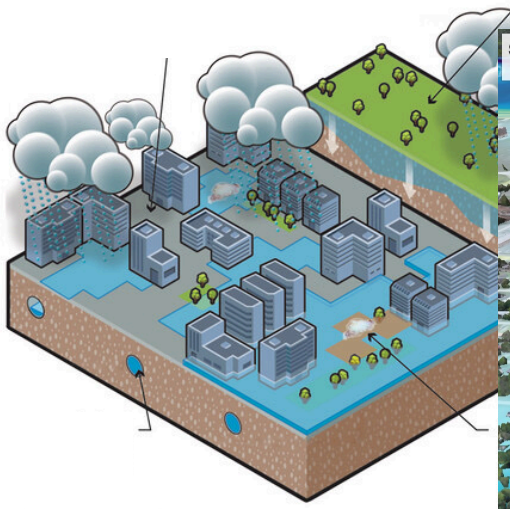
FLOOD INUNDATION MAPPING AND PREDICTION



=

Developed CREST-iMAP to Overcome These Limitations

Pluvial Flood



Fluvial Flood and Reservoir



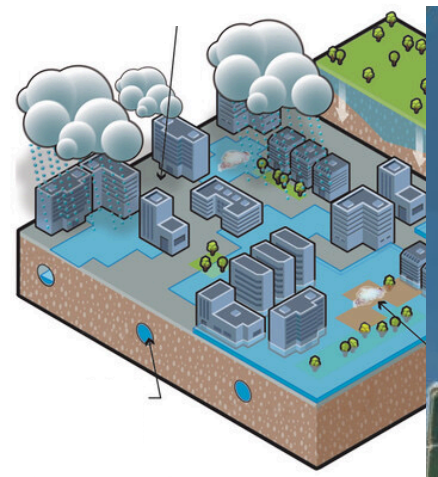
CREST-iMAP

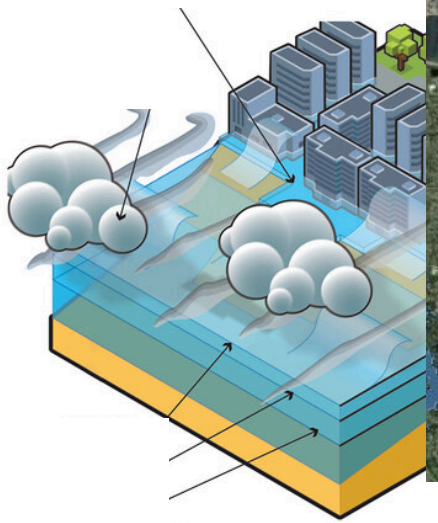


Developed CREST-iMAP to Overcome These Limitations

Pluvial Flood

Fluvial Flood and Reservoir



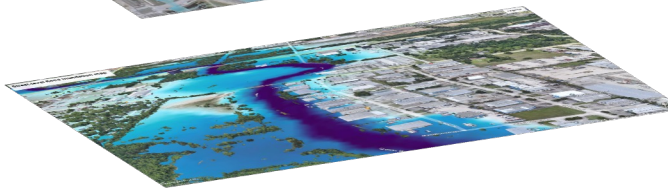




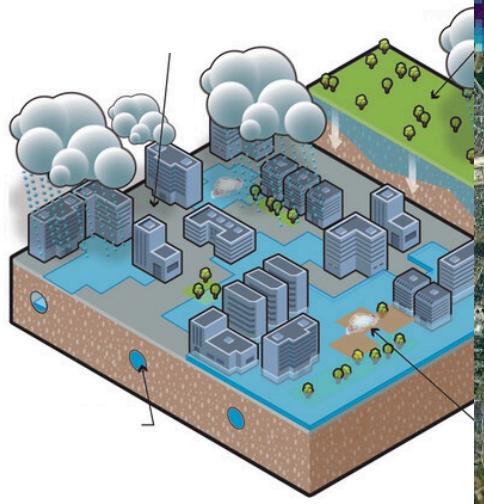
CREST-iMAP

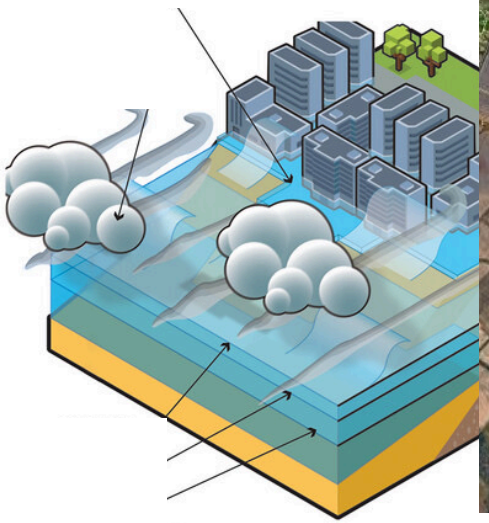
FLOOD INUNDATION MAPPING AND PREDICTION

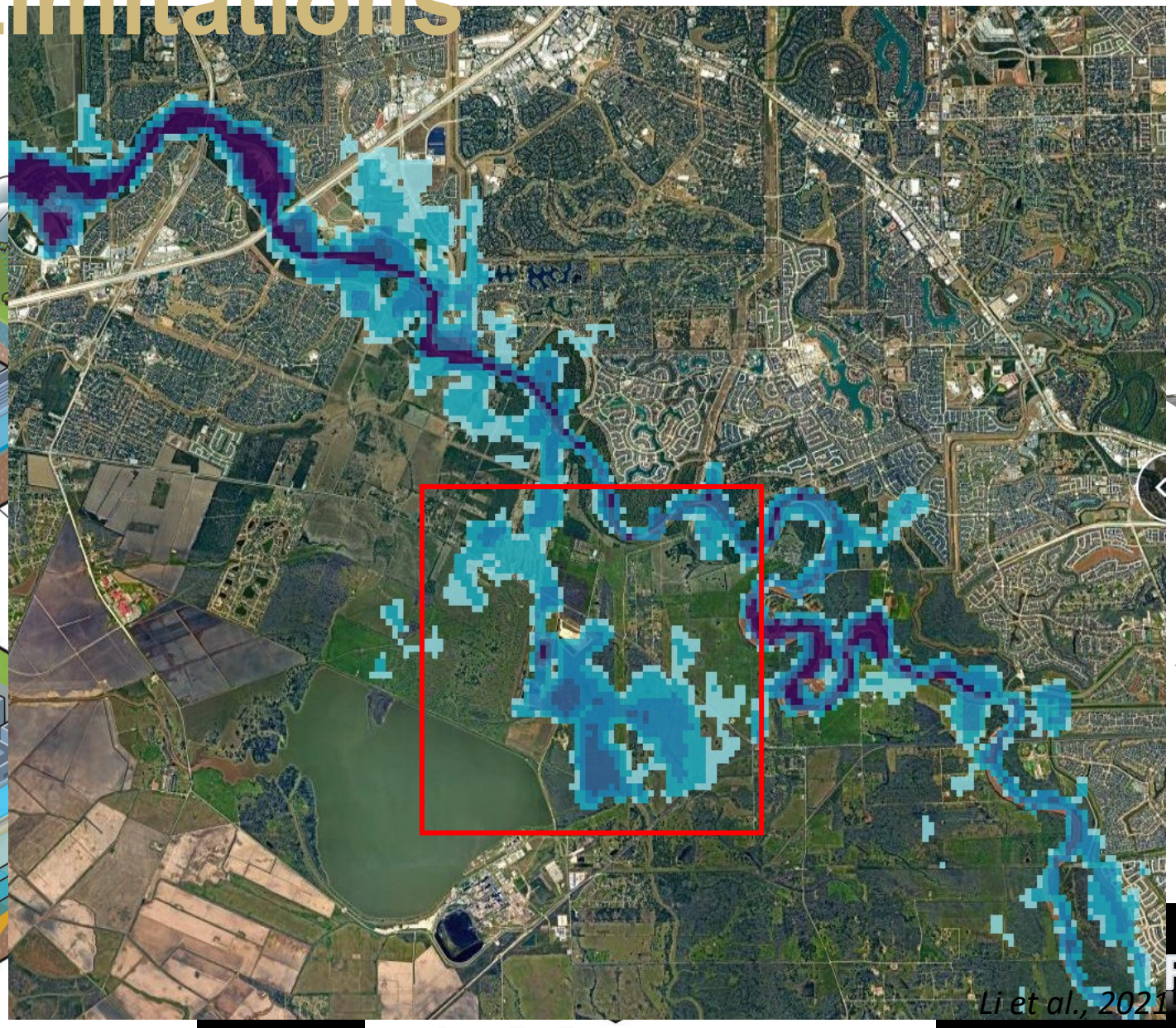




Developed CREST-iMAP to Overcome These Limitations

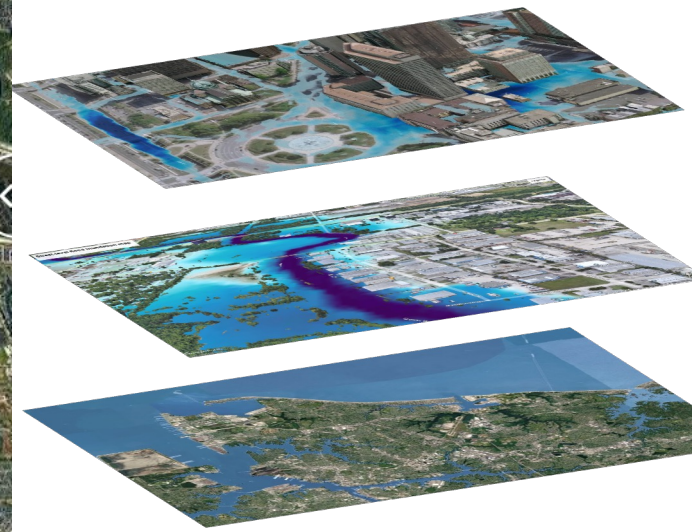






CREST-iMAP

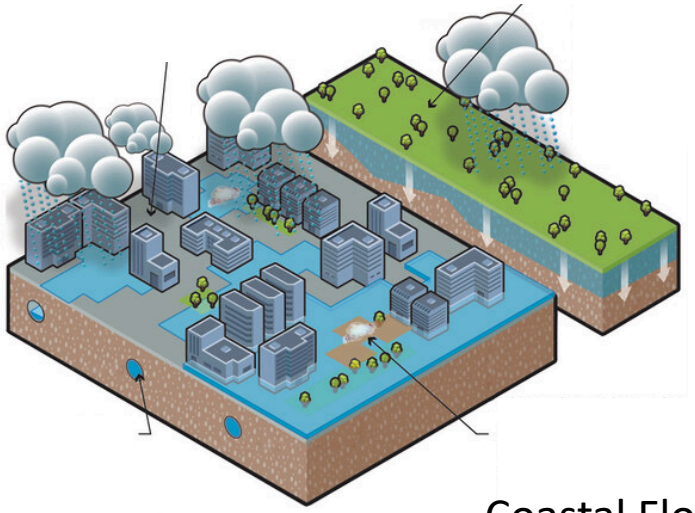
FLOOD INUNDATION MAPPING AND PREDICTION



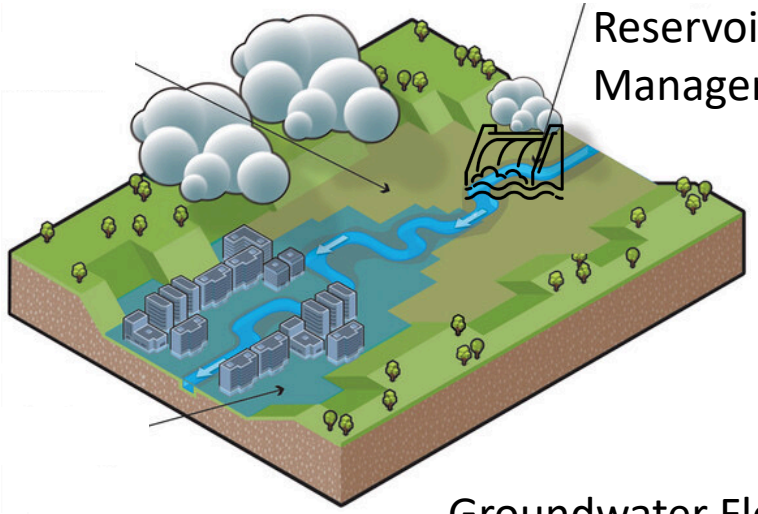
Li et al., 2021

Developed CREST-iMAP to Overcome These Limitations

Pluvial Flood

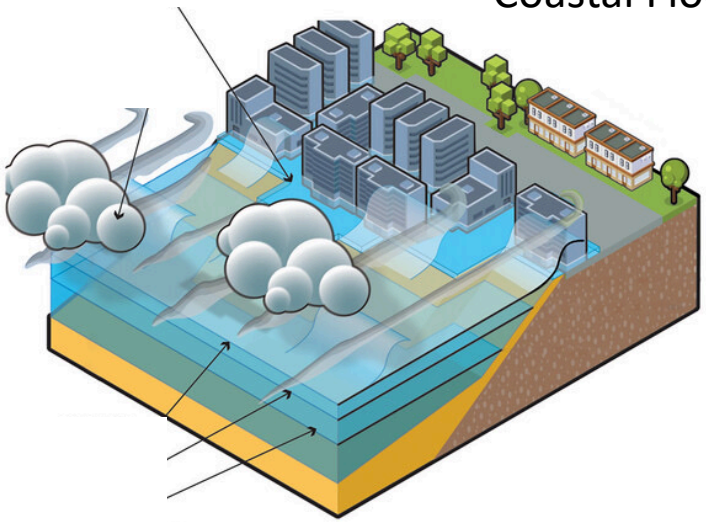


Fluvial Flood and Reservoir Management

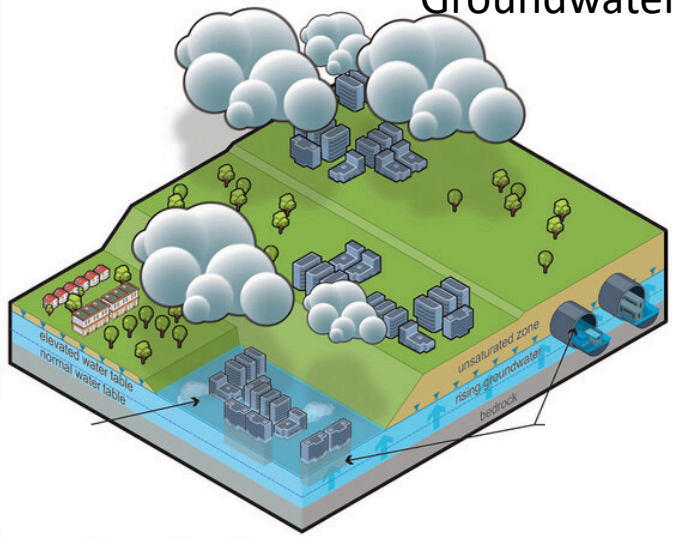


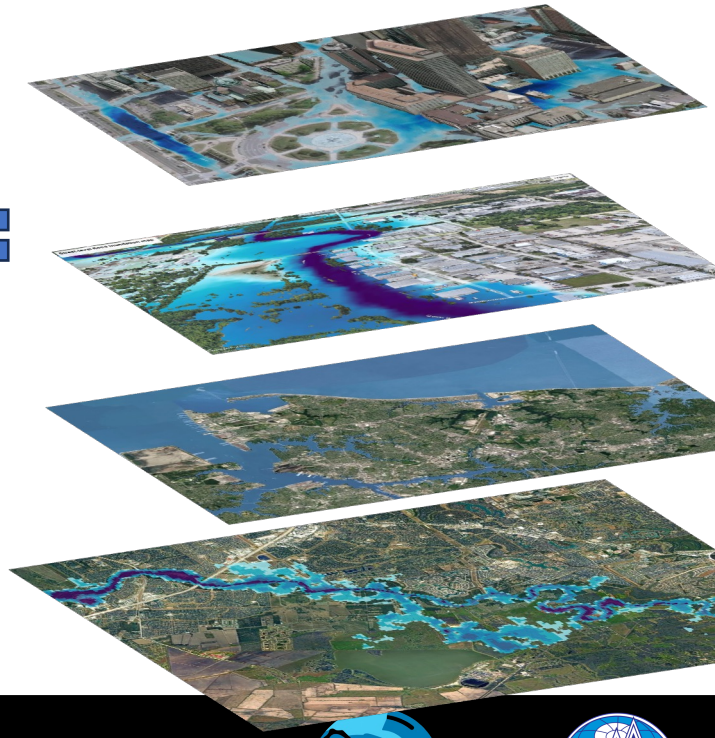
CREST-iMAP

Coastal Flood



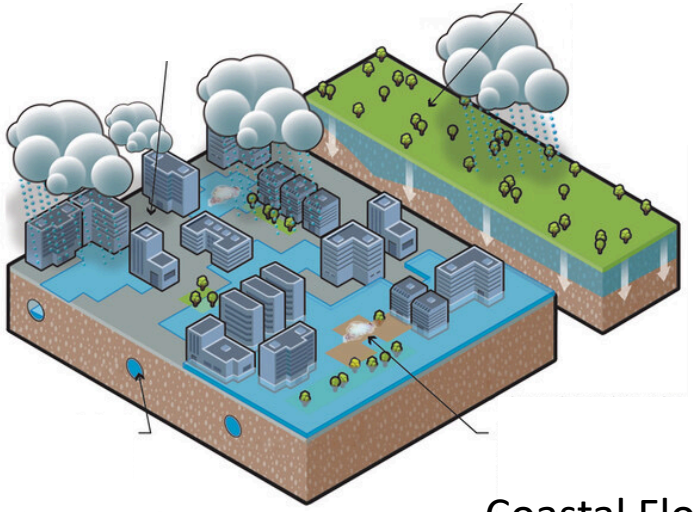
Groundwater Flood



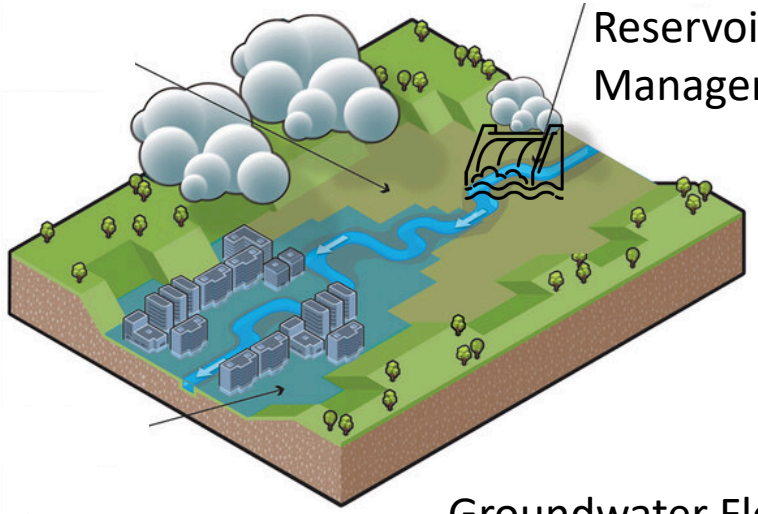


Developed CREST-iMAP to Overcome These Limitations

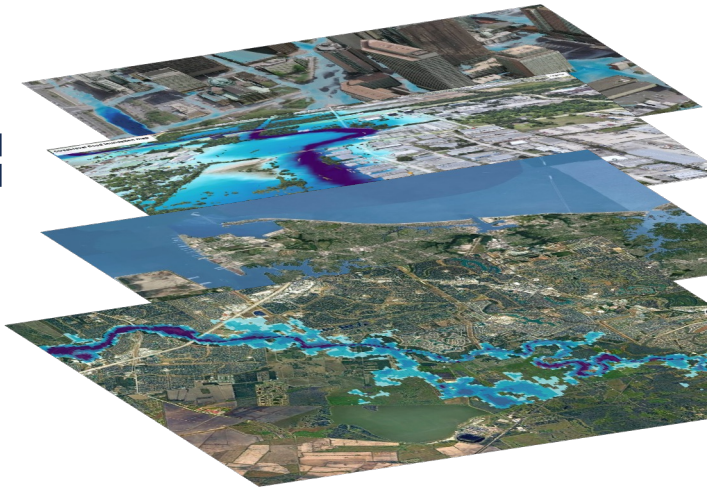
Pluvial Flood



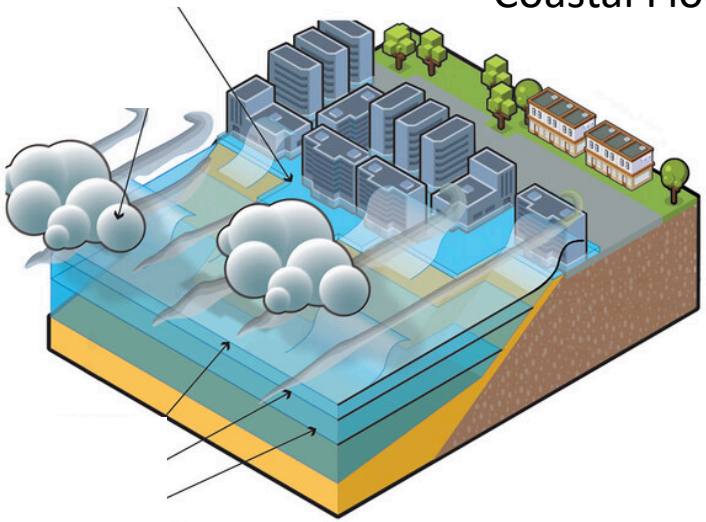
Fluvial Flood and Reservoir Management



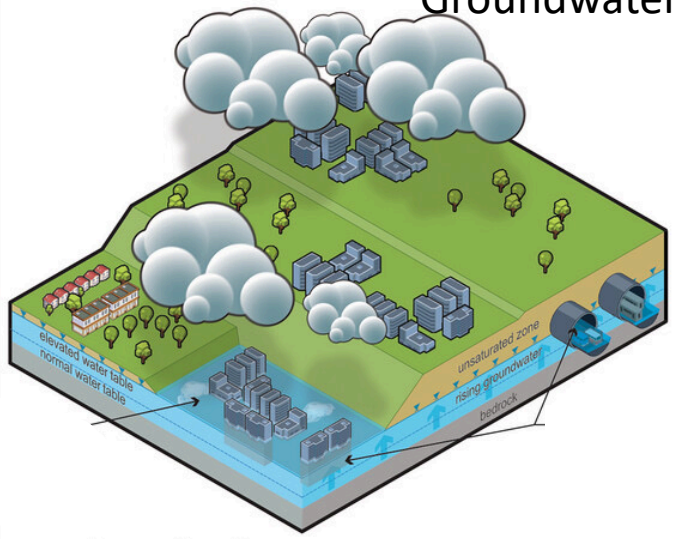
CREST-iMAP



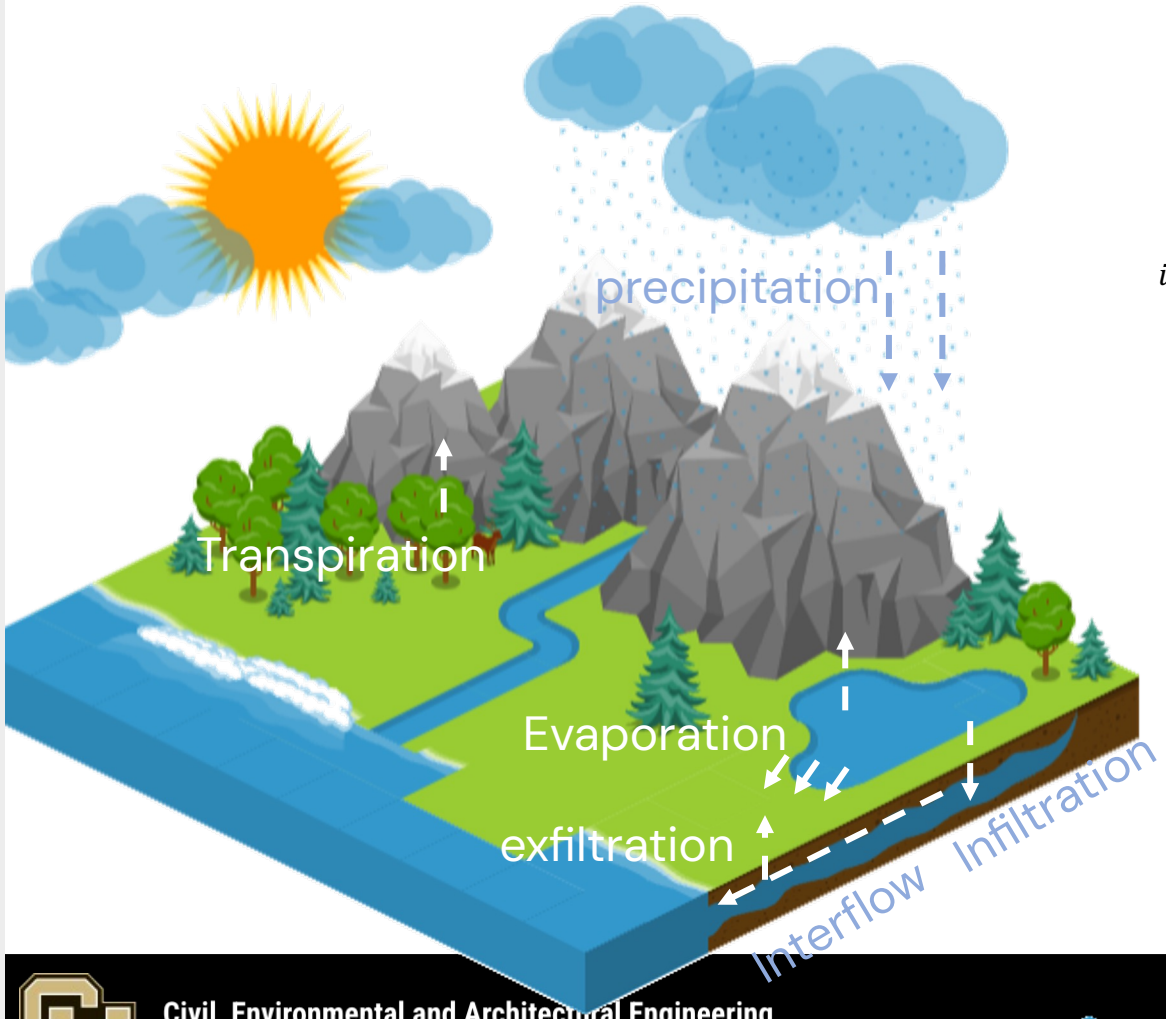
Coastal Flood

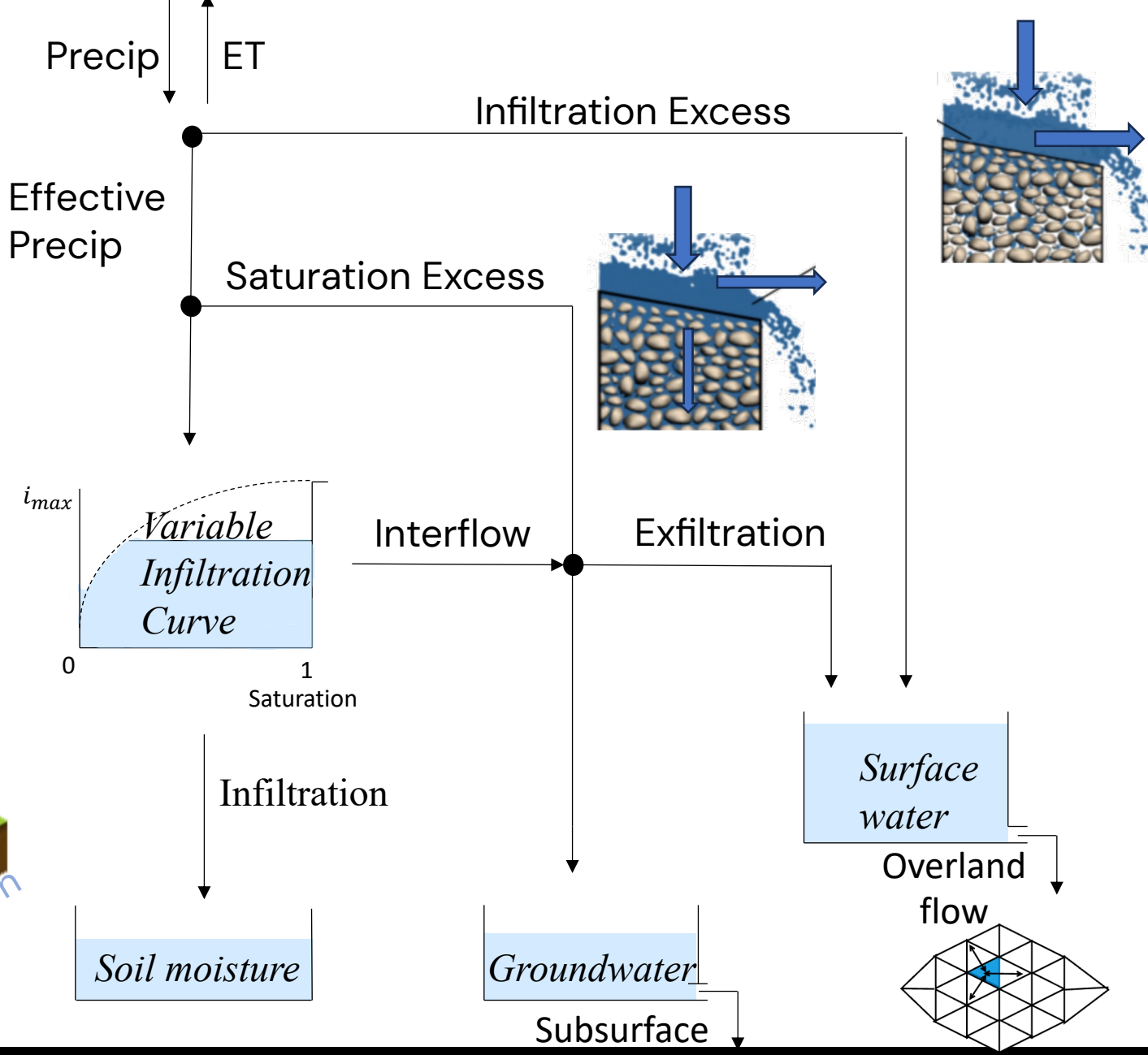


Groundwater Flood

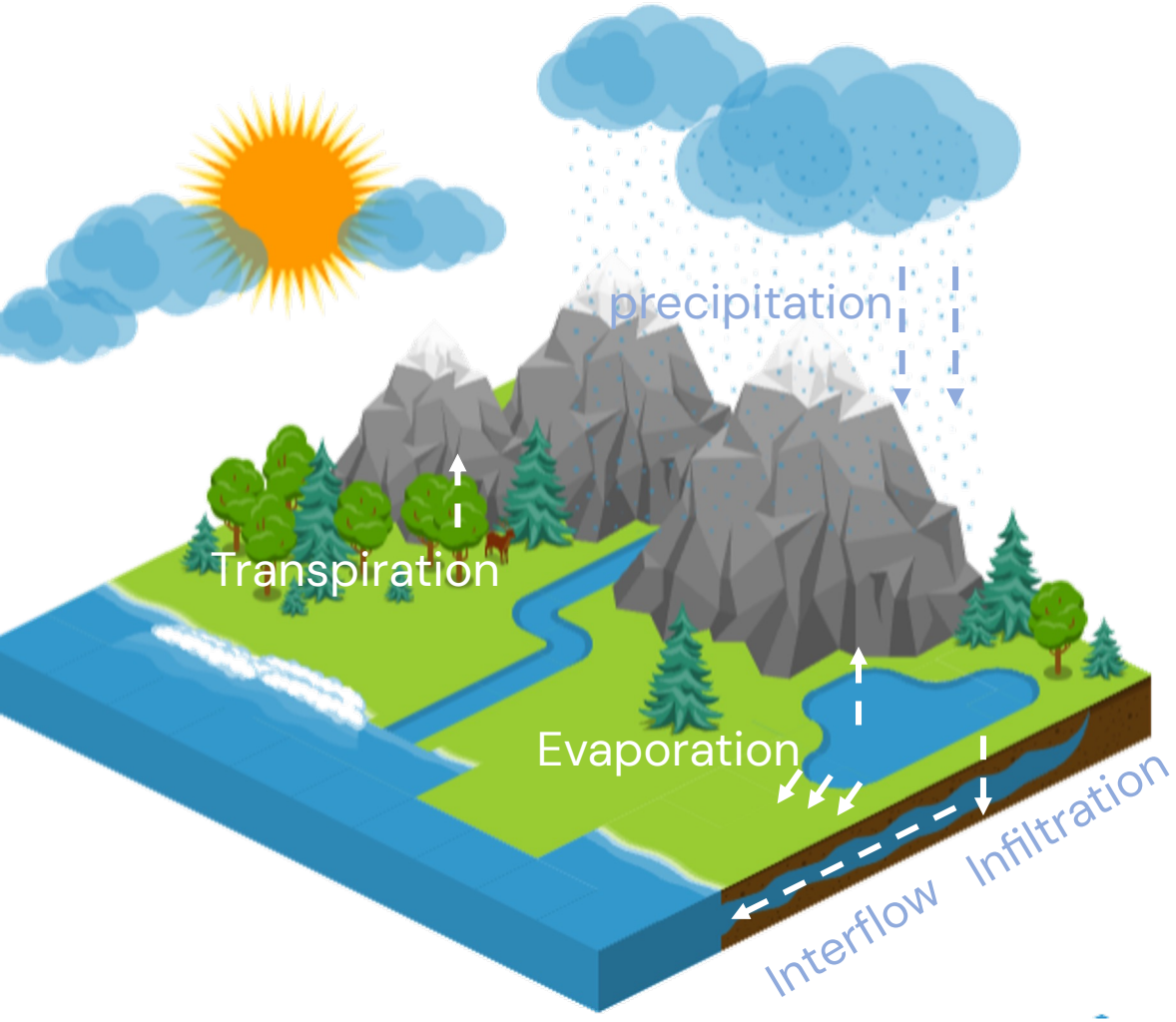


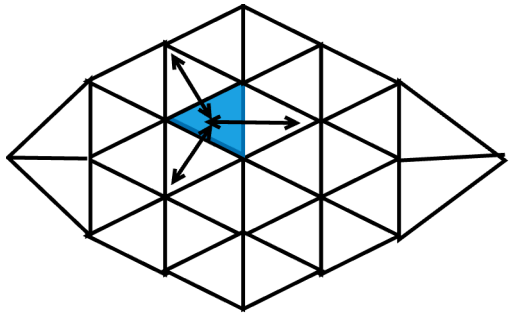
Processes in CREST-iMAP





Processes in CREST-iMAP





2D shallow water (mass and momentum balance)

Finite-volume numerical solution of governing PDE

Hyper-resolution (~ 1 m)

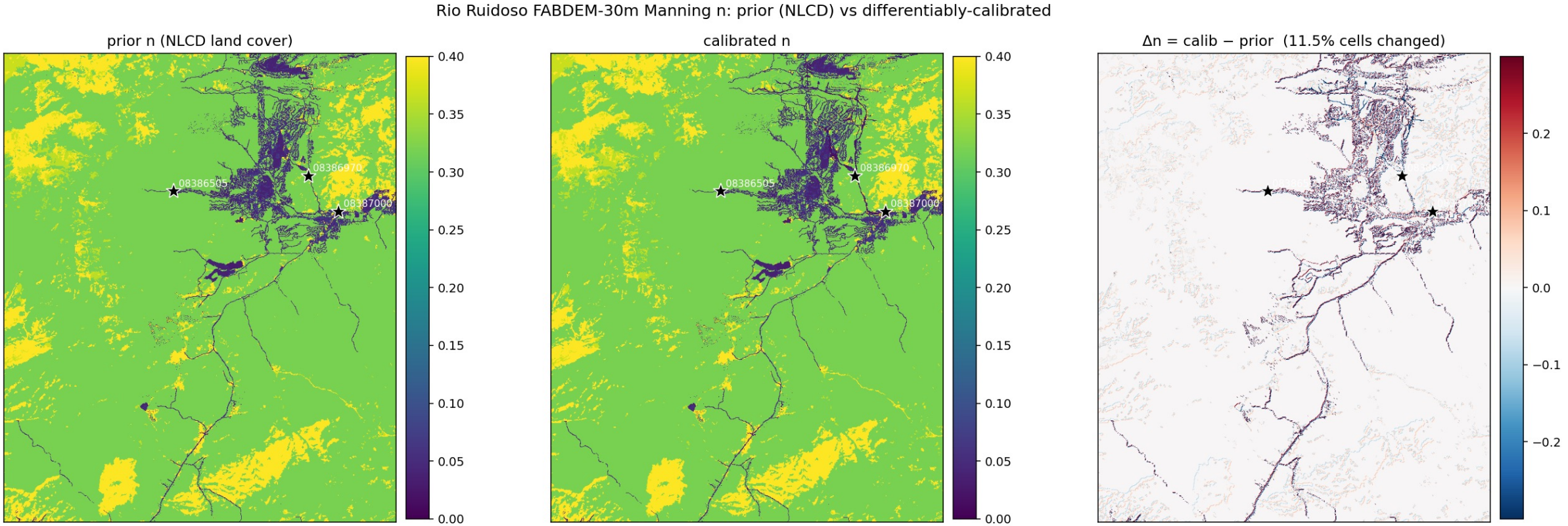
Semi-empirical groundwater model

Next gen FIM: Inunda

https://inunda.ai

Li, Z. (2026). Inunda: A GPU-Native, Agent-enabled, Differentiable Solver for High-Resolution Flood Inundation Modeling. *ArXiv*. https://arxiv.org/abs/2607.09614

Auto-diff calibration on manning's n



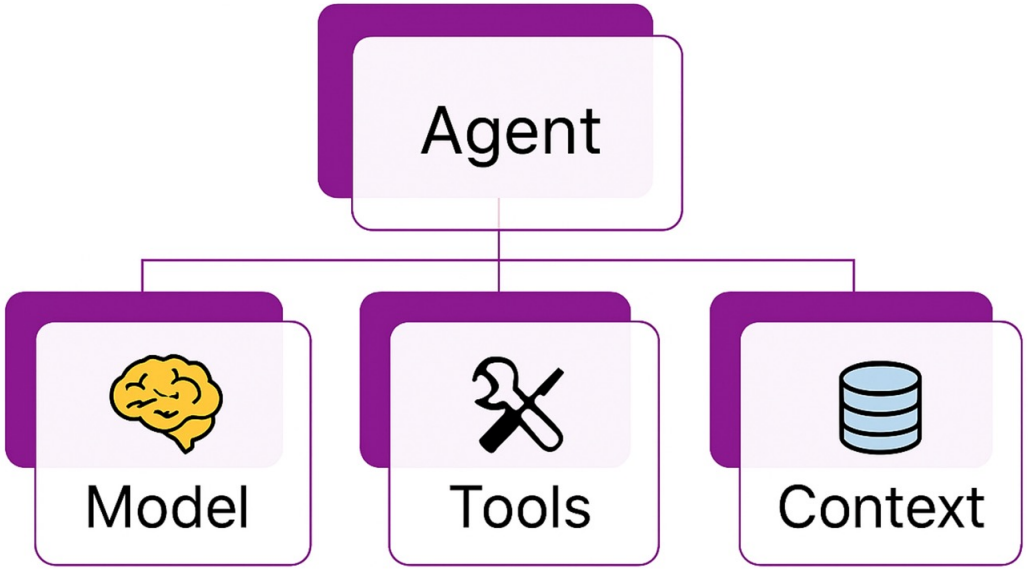
Agent-enabled flood modeling

Yan, S., Li, Z., Zhu, S., Wen, Y., Zhang, M., Chen, M., ... & Hong, Y. (2025, October). AQUAH: automatic quantification and unified agent in hydrology. In *2025 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)* (pp. 2947-2956). IEEE.

Li, Z., Yan, S., Cao, J., Zhang, M., Wei, A., Yoo, J., & Hong, Y. (2026). HydroAgent: Closing the Gap Between Frontier LLMs and Human Experts in Hydrologic Model Calibration via Simulator-Grounded RL. *arXiv preprint arXiv:2605.17792*.

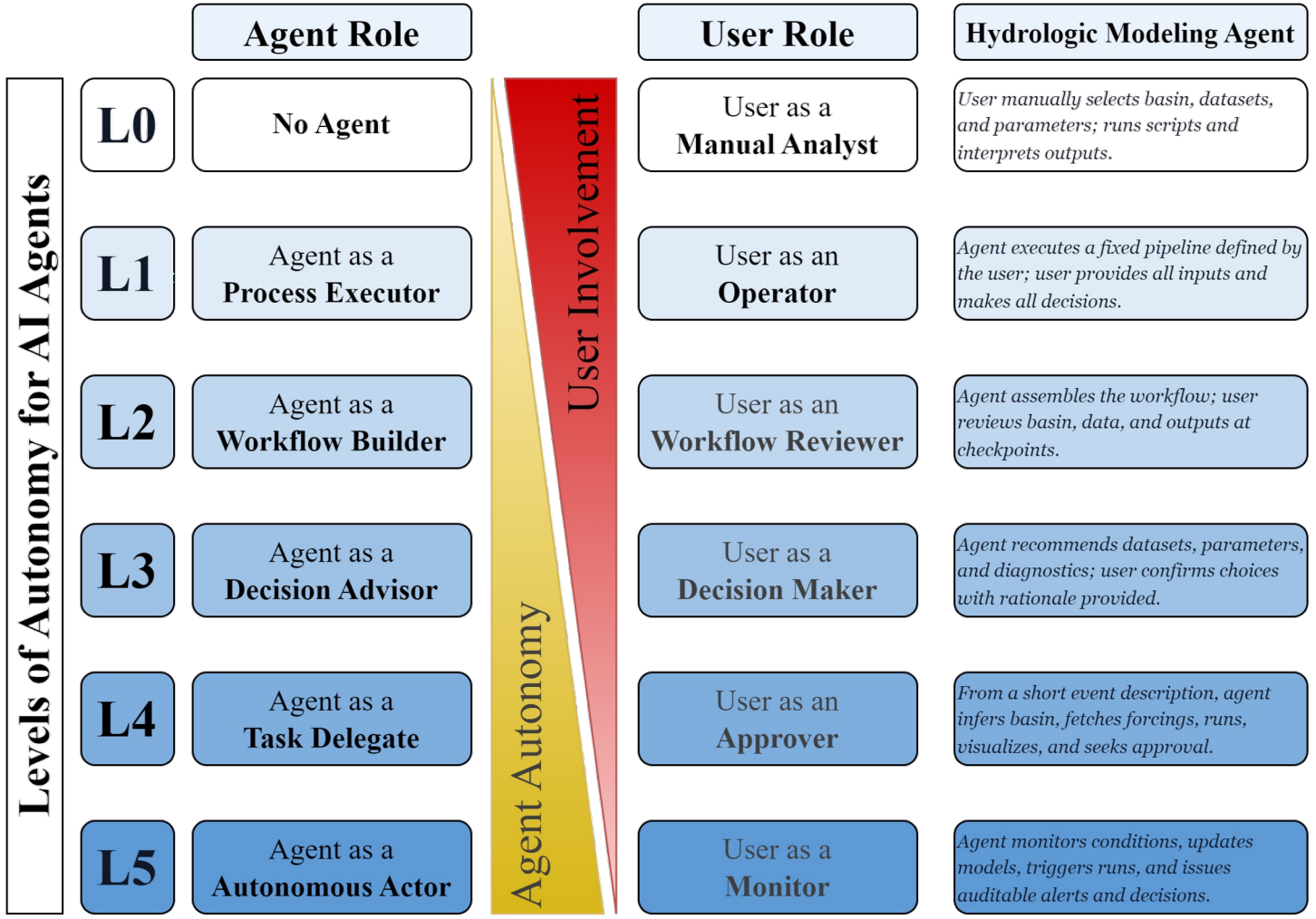
Yan, S., Chen, M., Li, Z., Wen, Y., Zhu, S., Zhang, M., ... & Hong, Y. (2026). AI agent for hydrologic modeling: Definition, development, and application. *Geophysical Research Letters*, *53*(13), e2025GL119814.

What are AI agents?



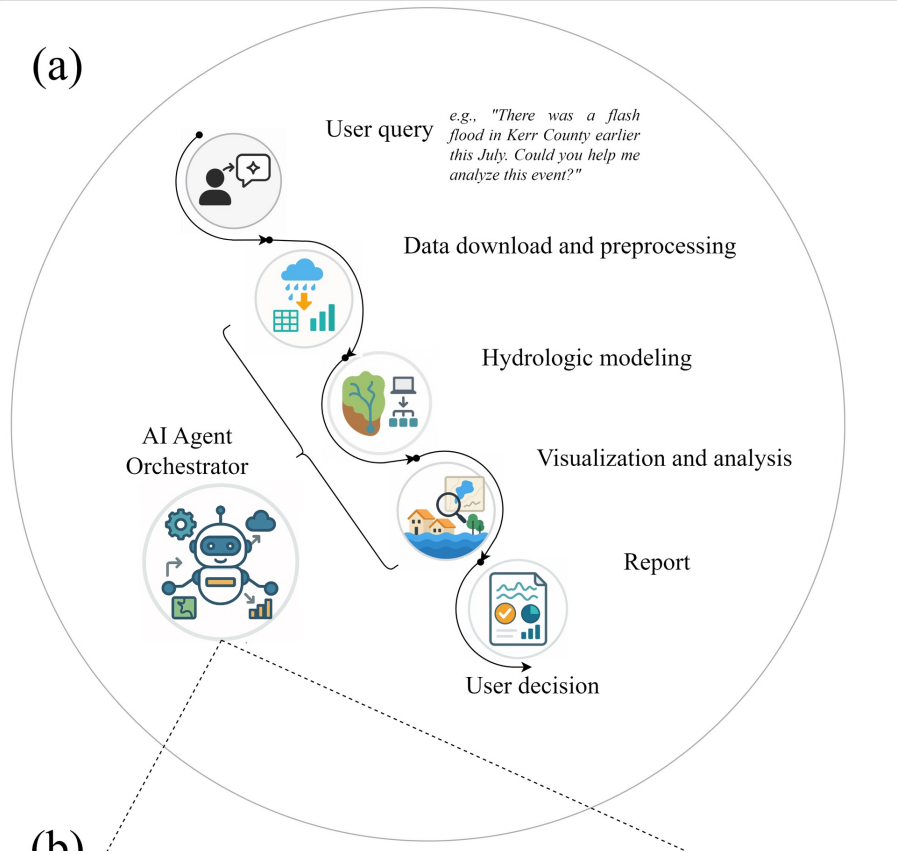
Coding agent: Claude code, Codex, Zcode, etc.

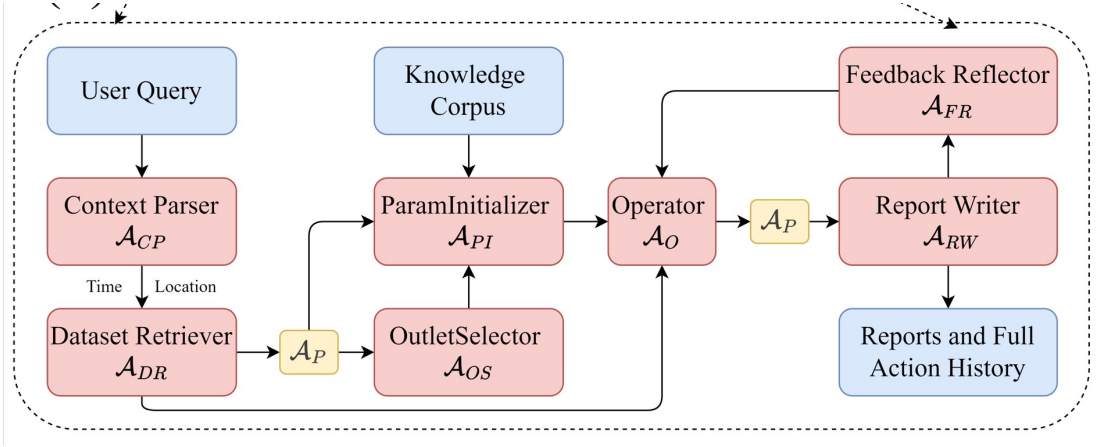
Levels of Automation in Hydro Modeling



Yan, Li, et al., 2025, ICCV, GRL

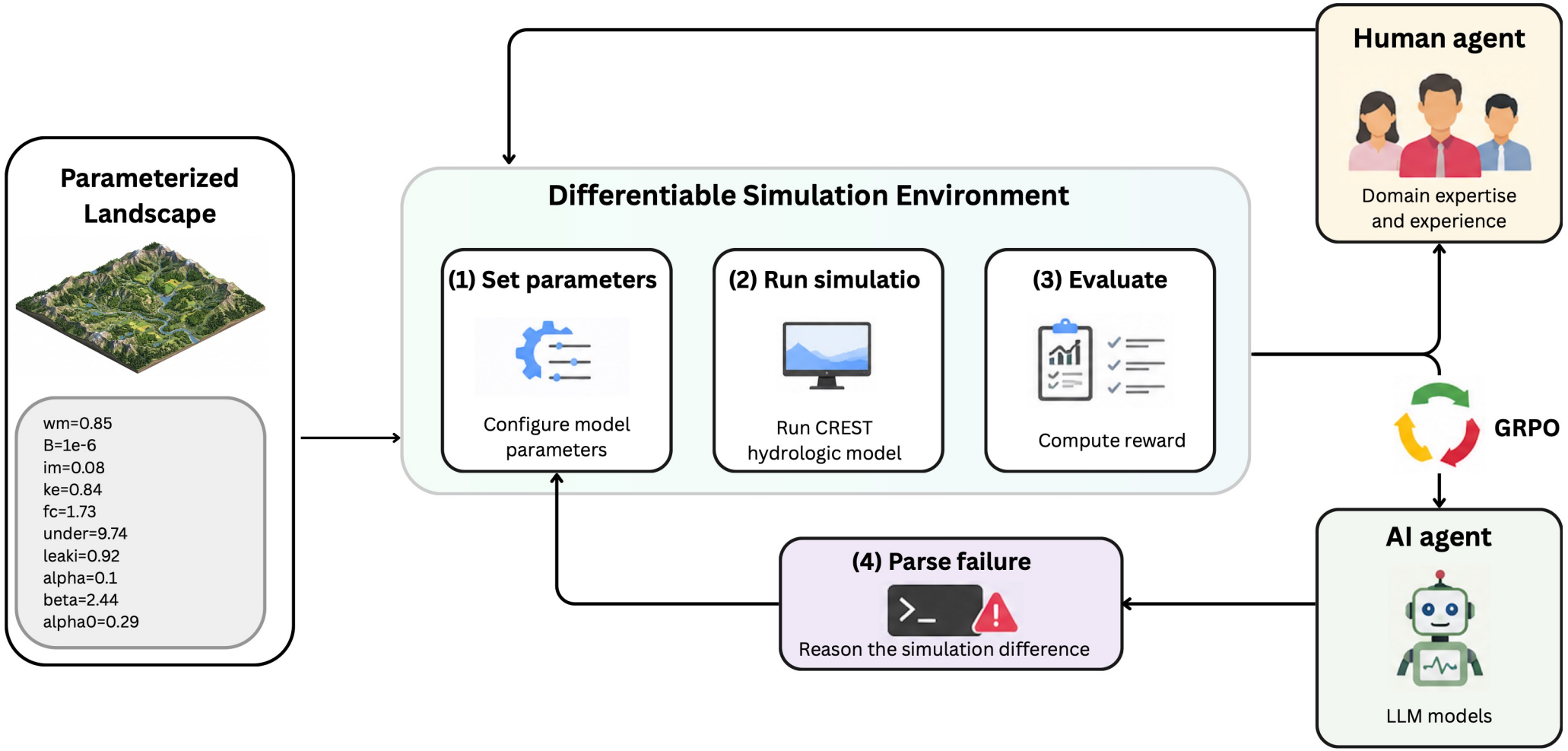
AQUAH: Automatic Quantification and Unified Agent in Hydrology





Yan, Li, et al., 2025, ICCV, GRL

HydroAgent: LLM with hydrologic knowledge

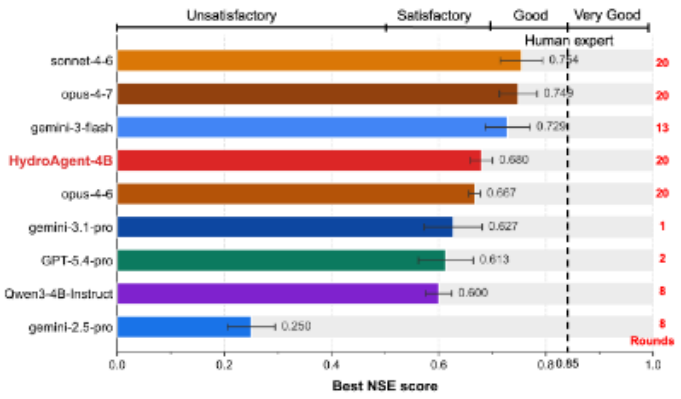


Li et al., 2026

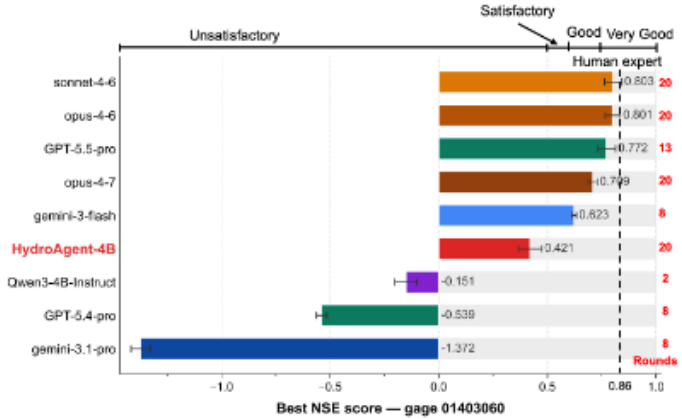
How does HydroAgent perform?

- Comparable performance with frontier models
- Better at long-horizon job

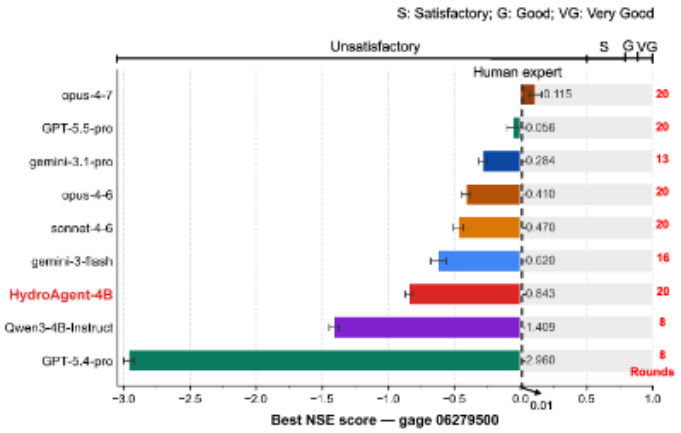
Gauge	Method	Best NSE	Sims	Turns	Parse fail
07144100	Baseline	0.34	15	50	0
	SFT-only	0.07	1	15	4
	HydroAgent	0.65	13	50	0
06279500	Baseline	−1.41	13	50	1
	SFT-only	−2.27	1	13	4
	HydroAgent	−0.84	17	50	0
02338660	Baseline	0.65	13	50	0
	SFT-only	−17.53	1	11	4
	HydroAgent	0.68	12	50	0
01403060	Baseline	−0.15	16	50	0
	SFT-only	0.58	4	16	4
	HydroAgent	0.40	14	50	0



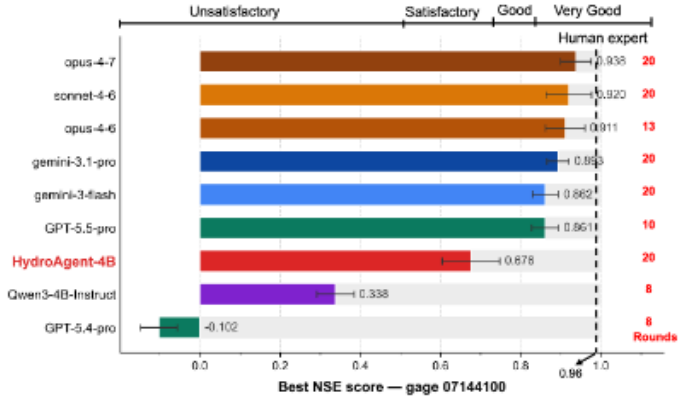
(a) 02338660



(b) 01403060

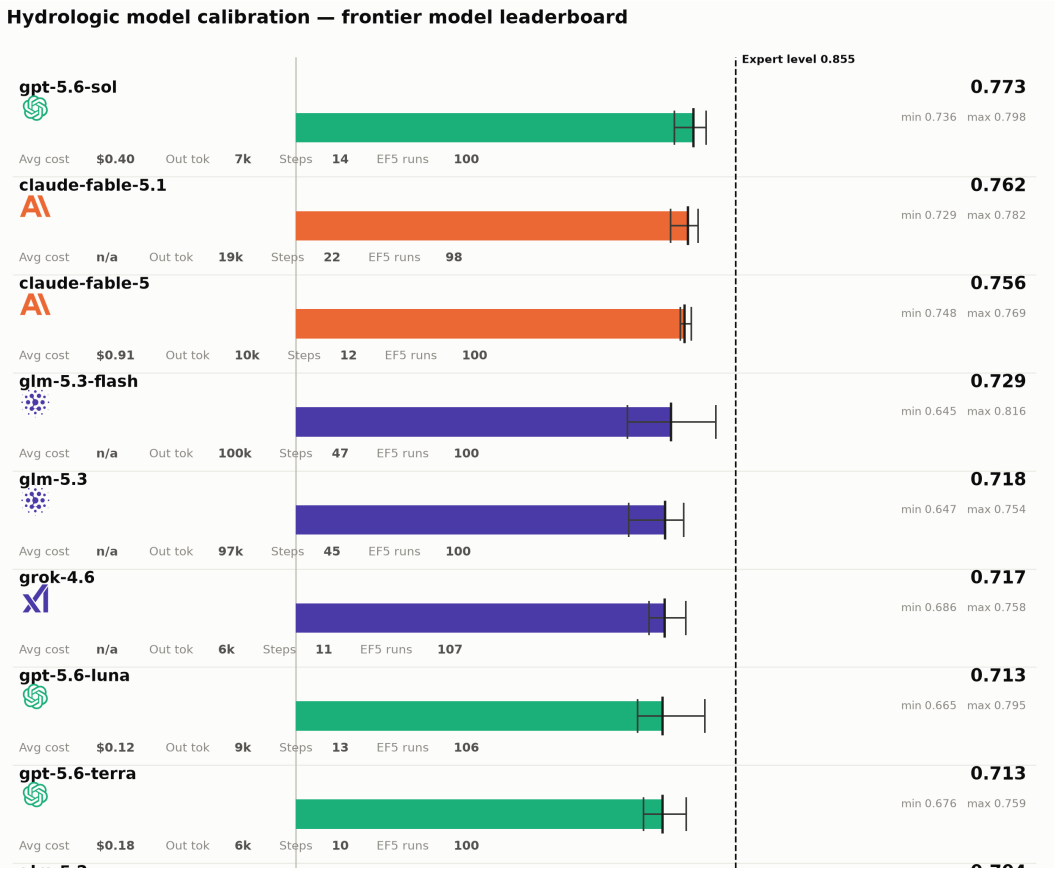


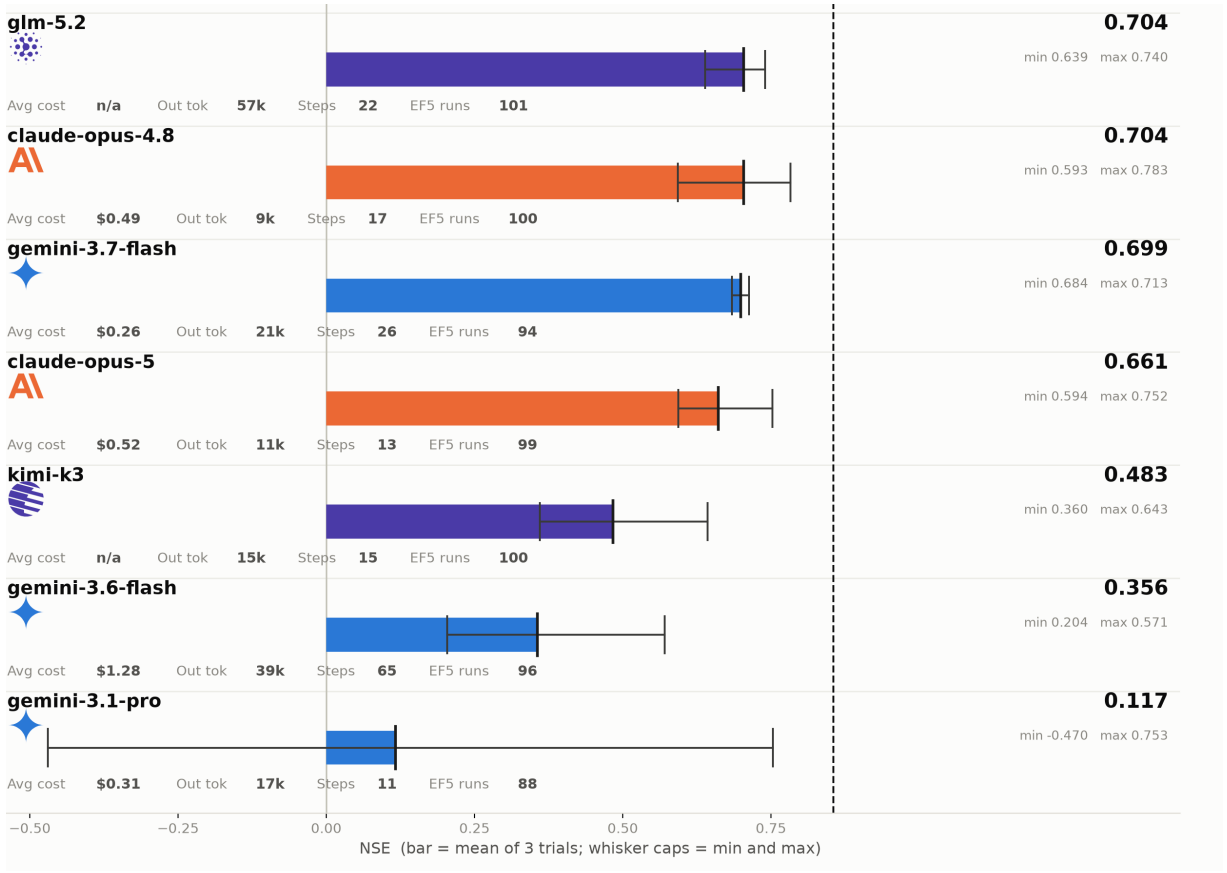
(c) 06279500



(d) 07144100

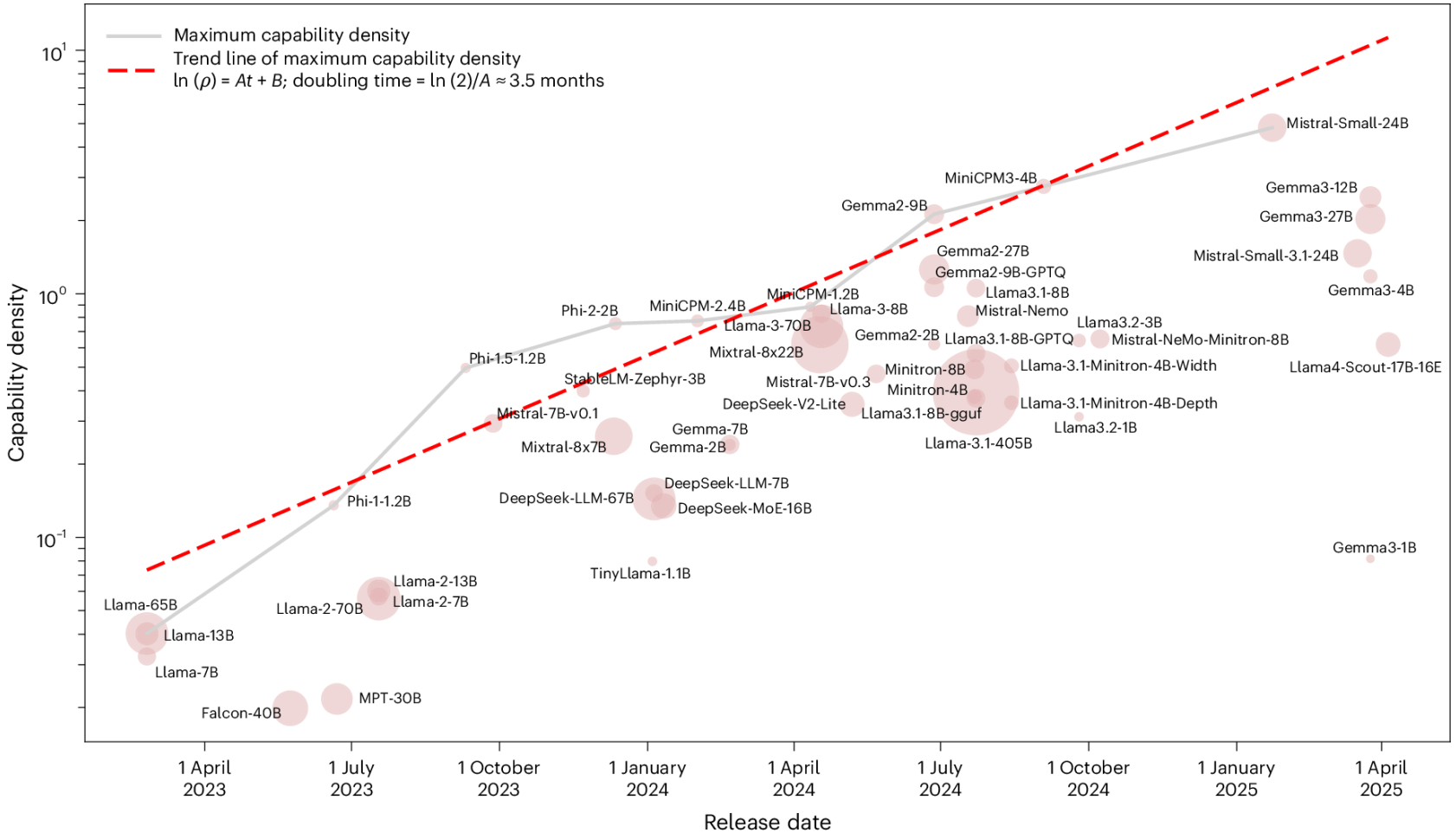
The hydro-model benchmark





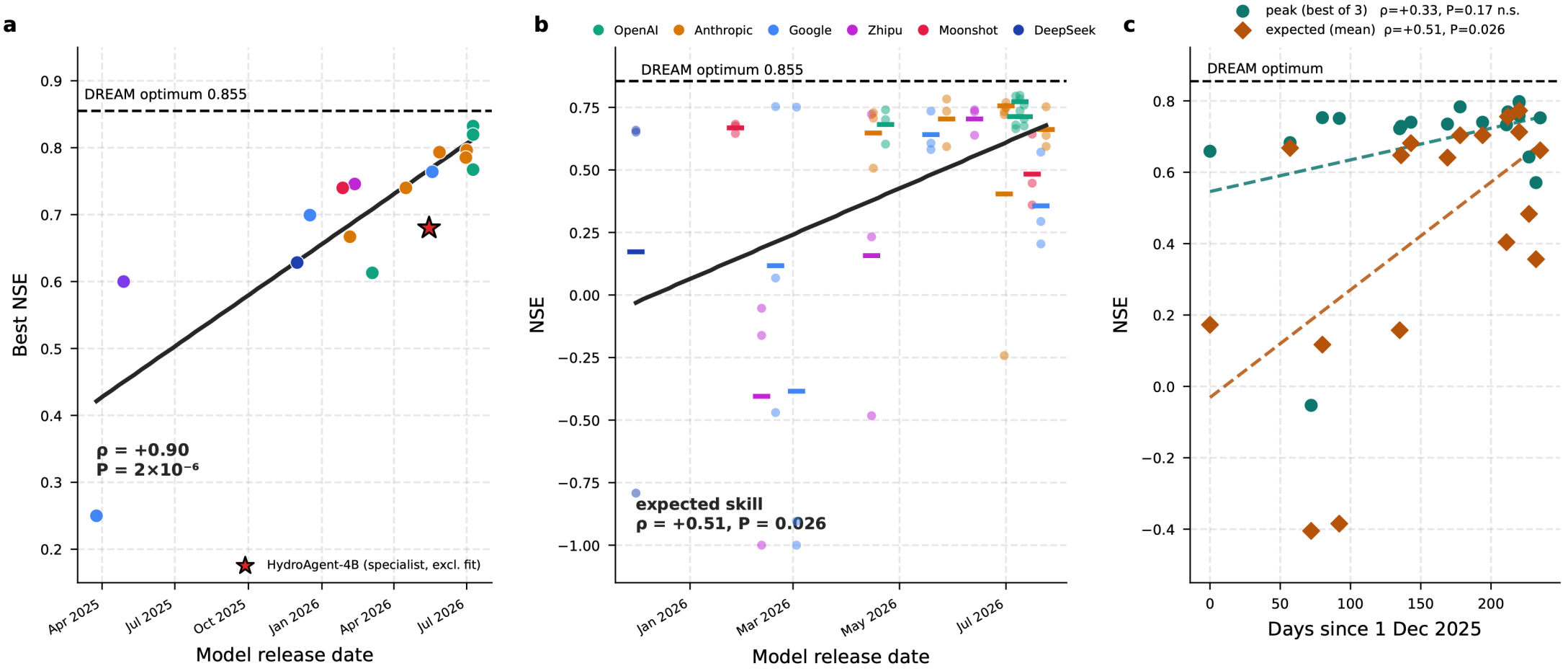
https://hydrors.us

Does the scaling law exist in hydro modeling?



Xiao et al., 2025

Does the scaling law exist in hydro modeling?



Prospects: Opportunities and Challenges

Challenges

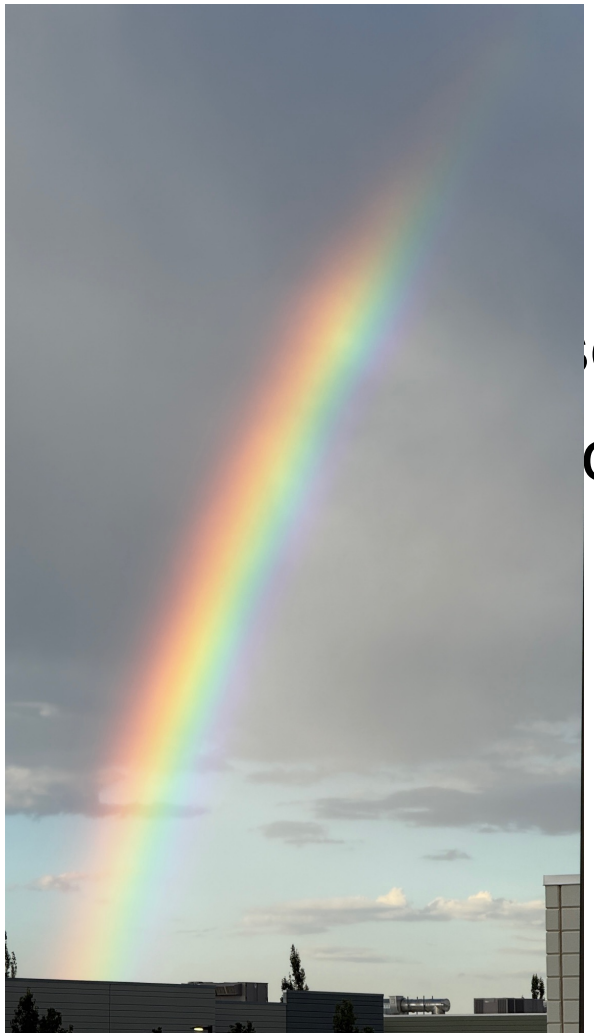
1. Most AI does not understand physics
   - Cannot extrapolate thru space and time
   - Cannot be generalized
2. Most AI does not understand causality
   - Sampled from large corpus of history

Opportunities

- From LLM to World Model
- Close the gap between simulation and decision making (L5 automation)

HydroTuring Initiative
水文图灵倡议

该二维码7天内(9月8日前)有效，重新进入将更新



earch is at an inflection point

odel could be the next wave of innovation

Questions?

Zhi.Li-2@Colorado.edu

X/Twitter: @chrimersslee

LinkedIn: https://www.linkedin.com/in/zhi-li-a79116167/

Xhs: